RAST: Reliability-Aware Spatiotemporal Modeling for Multiobject Tracking in Satellite Videos

Yuting Shi, *Graduate Student Member, IEEE*, Guanqun Wang, *Member, IEEE*, Yin Zhuang, *Member, IEEE*, Tong Zhang, *Member, IEEE*, and He Chen, *Member, IEEE*

***Abstract*—Satellite remote sensing videos enable persistent wide-area observation, yet multiobject tracking (MOT) in this domain remains challenging due to tiny targets, cluttered backgrounds, subtle and nonlinear displacements, and time-varying observation quality. Such reliability-variant conditions can induce temporal representation drift, intensify multitask interference between detection and motion estimation, and make conventional reliability-invariant association brittle, leading to fragmented trajectories and identity switches. From a unified reliability-aware spatiotemporal modeling perspective, we propose RAST, an online and causal tracking framework that progressively improves robustness from representation learning to data association. RAST consists of three complementary components: 1) a dual-view temporal enhancement mechanism (DVTEM) that couples forward inertia propagation with backward contextual verification within a causal window to suppress drift accumulation; 2) a motion-guided dynamic feature aggregation mechanism (MGDFAM) that constructs motion priors from a temporal feature bank and performs task-specific deformable attention with decoupled representations for detection and displacement estimation; and 3) a hybrid-driven cascade association strategy (HDCAS) that leverages detection confidence as a proxy of observation reliability to adaptively switch between displacement-guided matching and Kalman-filter-based inertial association. Extensive experiments on two satellite video MOT benchmarks, SatVideoDT and SatMTB-MOT, demonstrate that RAST consistently outperforms state-of-the-art (SOTA) methods. In particular, RAST achieves 67.0% MOTA and 75.1% Rcll on SatVideoDT, and attains the best MOTA across all three subsets of SatMTB-MOT, validating its robustness under crowded tiny-object scenarios and multiclass variations.**

***Index Terms*—Multiobject tracking (MOT), reliability-aware, satellite video, spatiotemporal modeling.**

I. Introduction

SATELLITE remote sensing videos provide a unique capability for persistent wide-area observation, enabling continuous monitoring of dynamic Earth-surface activities, such as traffic flow, maritime surveillance, and airport operation management [1]. Compared with still imagery, satellite videos contain rich temporal cues that are essential for understanding moving targets and their long-term behaviors [2]. As a fundamental component in practical remote sensing interpretation systems, multiobject tracking (MOT) in satellite videos aims to detect and associate multiple moving instances across frames, producing temporally consistent trajectories for downstream spatiotemporal analytics [3].

Despite rapid progress in general MOT, robust tracking in satellite videos remains challenging in real-world scenarios. Satellite platforms typically observe extremely large fields of view, while targets of interest (e.g., vehicles, ships, and airplanes) often occupy only a small number of pixels and exhibit weak appearance cues [4]. Meanwhile, complex backgrounds (e.g., roads, buildings, waves, and clouds) [5] and frequent target interactions (e.g., crossing, merging, and partial occlusion) [6] can induce severe ambiguities for both detection and association. These factors make satellite video MOT particularly vulnerable to missed detections, fragmented tracklets, and identity switches, especially under crowded small-object conditions [2]. Fig. 1 visualizes these characteristics on a representative satellite video sequence, where ground-truth trajectories confirm the subtle yet complex motion patterns of few-pixel targets.

As further evidenced in Fig. 1, satellite videos are characterized by extremely small objects and subtle but often nonlinear cross-frame displacements, which are tightly coupled with the unique acquisition conditions of spaceborne sensors. Unlike natural videos with relatively stable viewpoints and rich textures [7], satellite videos usually present low-contrast and low-resolution observations [8], subtle cross-frame displacements [9], and spatially heterogeneous background dynamics [10]. Such weak appearance cues and subtle motion signals make temporal representation learning and motion reasoning highly sensitive to background dynamics and transient degradations (e.g., occlusion and low contrast). Consequently, the observation quality of targets can vary significantly over time: a target may be clearly visible in some frames but become weak, noisy, or partially occluded in others. This reliability-variant nature of satellite observations poses a fundamental challenge to learning-based tracking systems, as temporal representations may drift, and motion cues may become unreliable under transient degradations.

From a reliability-aware spatiotemporal modeling (RAST) perspective, we identify three coupled limitations that hinder existing satellite video MOT approaches.

Received 1 February 2026; revised 6 May 2026; accepted 15 June 2026. Date of publication 18 June 2026; date of current version 30 June 2026. This work was supported in part by the General Program of National Natural Science Foundation of China under Grant 62371048, in part by Ye Qisun Science Foundation of the National Natural Science Foundation of China under Grant U2341202, in part by Beijing Institute of Technology Kunpeng and Ascend Center of Cultivation, and in part by the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing (SBIIP). *(Corresponding author: Guanqun Wang.)*

Yuting Shi, Guanqun Wang, Yin Zhuang, and He Chen are with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing (SBIIP), School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: wgq@bit.edu.cn).

Tong Zhang is with the School of Electronics, Peking University, Beijing 100871, China.

Digital Object Identifier 10.1109/TGRS.2026.3705110

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

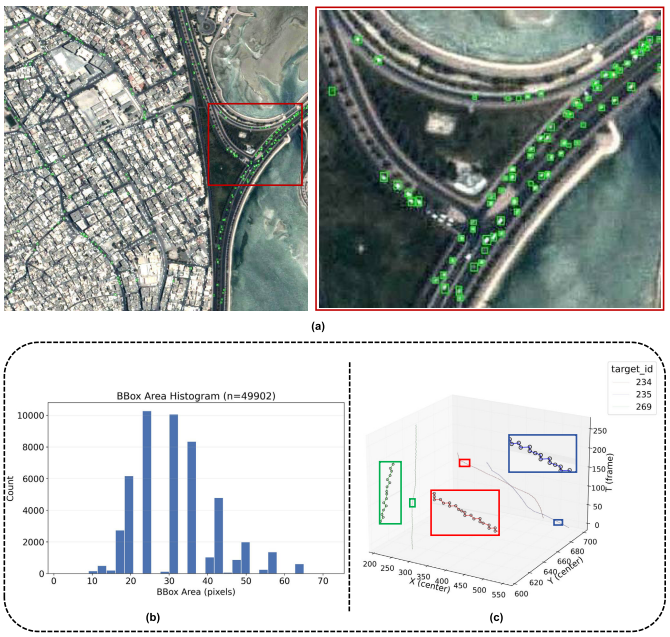


Fig. 1. Illustration of key challenges in satellite video MOT on SatVideoDT (video 001). (a) Representative frame with an enlarged region, showing extremely small targets embedded in cluttered backgrounds. (b) Distribution of ground-truth bounding-box areas over the sequence, demonstrating the extremely small-object property in satellite videos. (c) Ground-truth trajectories of three sampled targets over the whole sequence; the zoomed-in boxes highlight subtle yet nonlinear motion patterns.

First, temporal modeling is often insufficient to prevent representation drift. Many methods [11], [12], [13], [14], [15] adopt one-way temporal propagation or short-term aggregation, implicitly assuming that historical features remain reliable once encoded. However, under tiny targets and cluttered backgrounds, noisy historical cues can be recursively accumulated, leading to temporal representation drift and gradually degrading the discriminability of per-frame features. As a result, targets may be progressively suppressed, causing track fragmentation and reduced trajectory coverage.

Second, multitask learning in satellite video MOT is often built upon shared representations, which can induce task interference. A common design in existing trackers is to optimize detection and motion/displacement estimation on top of the same backbone features (or a shared intermediate embedding) for efficiency and end-to-end training [16], [17]. However, these objectives favor different feature characteristics: detection requires appearance-discriminative and background-suppressing representations, whereas displacement estimation prefers motion-sensitive cues that emphasize cross-frame changes [18]. For satellite videos with tiny targets and cluttered backgrounds, forcing both tasks to compromise in a single feature space often leads to a suboptimal tradeoff, degrading detection robustness and making motion cues less reliable for association.

Third, association mechanisms are typically single-mode and lack reliability-adaptive flexibility. Online association may rely on data-driven displacement predictions that can be agile for clear targets but become noisy under degraded observations [16] or rely on physics-driven inertial models that are stable but may lag behind nonlinear motions [19], [20], [21], [22]. Since motion cues themselves are reliability-variant in satellite videos, a fixed association strategy is often brittle, leading to either increased identity switches or reduced recall when observation quality changes.

Motivated by these observations, we propose to reinterpret satellite video MOT from a unified RAST perspective, aiming to simultaneously: 1) stabilize temporal representations under degraded observations; 2) mitigate task interference between detection and displacement estimation; and 3) perform reliability-adaptive association to improve long-term identity consistency.

In this article, we present RAST, a dual-view temporal enhancement and motion-decoupled tracking framework for robust MOT in satellite videos. RAST follows a progressive three-stage design and operates online, processing each frame in a causal manner without future frame information. *First*, we introduce a *dual-view temporal enhancement mechanism* (DVTEM) that couples forward motion inertia propagation with backward contextual verification within a causal window, effectively suppressing temporal drift by rectifying unreliable historical cues. *Second*, we design a *motion-guided dynamic feature aggregation mechanism* (MGDFAM) to construct motion priors from a temporal feature bank and perform task-specific deformable attention with *decoupled queries and value features* (i.e., using independent projections and task-dependent value streams) for detection and displacement estimation, thereby alleviating task interference. *Third*, we propose a *hybrid-driven cascade association strategy* (HDCAS) that uses detection confidence as a practical proxy of observation reliability, enabling reliability-adaptive switching between displacement-guided matching and Kalman-filter-based inertial association as a safety net.

Extensive experiments on two representative satellite video MOT benchmarks, *SatVideoDT* [23] and *SatMTB-MOT* [24], demonstrate the effectiveness and robustness of RAST. On the challenging SatVideoDT dataset, our method achieves state-of-the-art (SOTA) performance with 67.0% MOTA, 75.1% Rcll, and 884 mostly tracked trajectories (MT) while reducing the number of mostly lost trajectories (ML) to 137. On the multicategory SatMTB-MOT dataset, RAST consistently outperforms existing approaches across all object categories, with MOTA scores of 63.0%, 56.1%, and 73.4% for vehicles, ships, and airplanes, respectively. Comprehensive ablation studies validate the contribution of each component, with the dual-view temporal enhancement and motion-guided decoupling showing particularly strong complementary effects.

The main contributions of this work are summarized as follows.

1) We reinterpret satellite video MOT from a unified RAST perspective and reveal three coupled bottlenecks in existing approaches, including temporal representation drift, multitask interference, and reliability-invariant data association.
2) We propose RAST, an online and causal three-stage tracking framework that progressively enhances temporal representations, learns motion-decoupled features for detection and displacement estimation, and performs reliability-adaptive cascade association for robust MOT in satellite videos.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

3) We develop a DVTEM that couples forward inertia propagation with backward contextual verification within a causal window, effectively suppressing drift accumulation under degraded observations.
4) We design an MGDFAM and an HDCAS: the former performs task-specific deformable attention with decoupled queries and value features to mitigate task interference, while the latter leverages confidence-based reliability stratification to switch between displacement-guided and Kalman-filter-guided matching for robust association.

II. Related Works

A. Satellite Video MOT

Existing approaches to satellite video MOT largely follow two main paradigms adapted from natural video tracking: tracking-by-detection (TBD) and joint-detection-and-tracking (JDT) [25]. TBD methods first detect objects in each frame and then link these detections online using appearance cues, motion models, and assignment algorithms [26]. In contrast, JDT methods perform detection and motion/identity feature learning in a unified network, aiming for higher efficiency and end-to-end optimization [27]. While these paradigms provide a convenient starting point, their performance can degrade noticeably in satellite videos due to the severe domain gap between spaceborne observations and natural video data [28].

To better handle satellite-specific challenges, recent works introduce dedicated designs for small-object representation enhancement, motion cue extraction, and association robustness [2]. Typical directions include leveraging temporal cues to stabilize features, exploiting motion priors under subtle displacements, and tailoring association strategies to crowded scenes and weak observations [11], [14], [29]. Nevertheless, robust tracking in satellite videos remains nontrivial: tiny targets with limited pixel support and cluttered backgrounds can still cause unreliable detections and motion estimates, leading to fragmented trajectories and identity switches under reliability-variant observations [30]. These limitations motivate RAST specifically designed for satellite video MOT.

B. Temporal Modeling and Drift Mitigation

Temporal modeling is a key ingredient in satellite video MOT, where per-frame appearance cues are often weak due to limited pixel support and cluttered backgrounds, making temporal cues essential for stabilizing target representations and improving tracking continuity [2]. In satellite videos, existing trackers commonly address temporal modeling and drift mitigation via two complementary strategies: 1) explicitly modeling cross-frame relations to capture higher-order temporal dependencies and 2) enhancing cross-frame feature updating and fusion to improve temporal consistency and reduce error accumulation under noisy cues and background pseudomotion.

Some works model cross-frame relations explicitly, e.g., TGraM [31] formulates tracking as an online graph reasoning process to mine high-order correlations across frames, and GMFTracker [32] reduces tiny-object detection drift by correcting classification–localization feature misalignment and leveraging graph-based relation reasoning.

Other approaches focus on cross-frame feature enhancement and fusion. For example, CFTracker [33] introduces cross-frame feature updating and training to learn temporal semantics and time-consistent motion prediction; MP^{2}Net [11] integrates mask propagation and motion prediction for tiny-target enhancement; FMA-Net [29] exploits flow-driven pixel-level motion priors to guide cross-frame feature fusion; SPAX [14] proposes hierarchical spatiotemporal fusion with symmetric/asymmetric cross-frame interactions; and DeTracker [34] decouples global platform motion from local object motion and performs cross-frame temporal fusion to improve trajectory stability in unstabilized satellite videos. In addition, PIFTrack [35] models target uncertainty with points of interest to mitigate localization drift under noisy annotations and adopts a diffusion-based ordinary differential equation model to capture nonlinear motions. Beyond recurrent and attention-based designs, recent works have also explored state-space models as an efficient alternative for temporal feature modeling in remote sensing video tracking [36], demonstrating strong capability in long-range dependency modeling with linear computational complexity.

Despite the above progress, effective temporal modeling and drift mitigation for satellite video MOT remain challenging. Many existing temporal modeling designs still tend to treat historical features as reliable once encoded; however, under time-varying observation quality, noisy cues can be accumulated over time and lead to temporal representation drift. Moreover, subtle cross-frame displacements and background dynamics (e.g., clutter and pseudomotion) can further corrupt temporal updates, making the learned temporal states increasingly biased and less discriminative. Therefore, satellite video MOT calls for drift-aware temporal modeling mechanisms that not only exploit motion inertia but also incorporate contextual verification from current observations to suppress error accumulation under reliability-variant conditions, yielding temporally stabilized representations for subsequent detection and association.

C. Motion Modeling and Data Association

Data association is a central component in MOT, whose goal is to link framewise detections into identity-consistent trajectories under occlusions and target interactions [20], [21]. In most TBD pipelines, association is formulated as a gated bipartite matching problem between the existing track set and current-frame detections, where motion prediction provides a prior for candidate filtering and the matching cost is typically defined by a combination of geometric consistency (e.g., IoU or center distance) and learned affinities [21].

A representative family relies primarily on geometry and motion priors. SORT [21] combines Kalman-filter-based state prediction with IoU-based matching, offering an efficient and widely used baseline for online association. To improve robustness beyond pure geometry, DeepSORT [20] further incorporates a learned appearance metric for matching, which reduces identity switches when targets are partially occluded or closely spaced.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

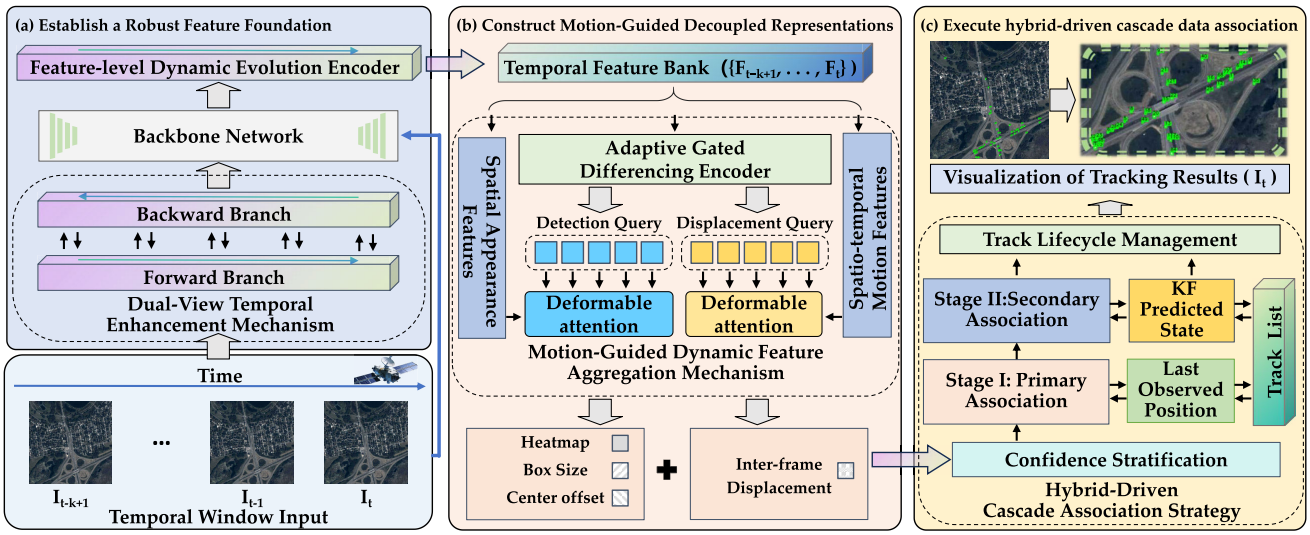


Fig. 2. Overall architecture of RAST. (a) Dual-view temporal enhancement with feature-level dynamic evolution encoding. (b) Motion-guided decoupled feature aggregation via an adaptive gated differencing encoder and dual-query deformable attention for detection and displacement prediction. (c) Hybrid-driven cascade association with confidence stratification.

Another practical direction improves fault tolerance by explicitly exploiting detection confidence. ByteTrack [22] performs score-stratified association and leverages low-score detections as auxiliary candidates to recover missed targets, which effectively alleviates trajectory fragmentation in crowded scenes.

From the motion-modeling perspective, Kalman-filter-based trackers have been continuously refined to better handle occlusions and nonlinear dynamics. OC-SORT [37] adopts an observation-centric strategy to reduce error accumulation during occlusion periods, while SOKF [38] introduces a sparse-observation Kalman filtering formulation with learned nonlinear state transitions to improve robustness under unreliable or missing measurements.

Beyond TBD pipelines with explicit matching, some methods follow the JDT paradigm by learning cross-frame correspondences (e.g., offsets or identity cues) together with detection in a unified network, which can further facilitate online association. For example, CenterTrack [16] regresses object centers and their cross-frame offsets to associate instances online, thereby using learned motion cues to guide matching. More recently, MOTIP [7] reformulates association as in-context identity prediction by decoding identity labels for current detections from historical trajectories.

In the satellite domain, association becomes particularly challenging due to extremely weak displacements, cluttered backgrounds, and frequent close-proximity interactions. Recent satellite-oriented trackers, therefore, tailor motion cues and association strategies to improve robustness under tiny-object conditions [11], [14], [29], where SPAX [14] combines hierarchical spatiotemporal fusion with Gaussian-based trajectory association to mitigate observational drift and accumulated errors.

Nevertheless, robust association in satellite videos remains challenging under reliability-variant observations [2], [14]. Learned motion cues (e.g., displacement/flow prediction) can be agile under clear observations but may become noisy when targets are low-contrast, partially occluded, or heavily interfered with by background clutter, leading to mismatches and identity switches. In contrast, physics-driven predictors (e.g., Kalman filtering) provide stable inertial estimates under degraded measurements but often lag behind nonlinear motions such as sudden turns and stop-and-go behaviors. Therefore, satellite video MOT calls for reliability-aware association mechanisms that can adaptively leverage complementary motion sources according to observation quality, thereby improving both robustness and long-term identity consistency.

III. Methodology

A. Overall Framework

As shown in Fig. 2, RAST is a dual-view temporal enhancement and motion-decoupled tracking framework for robust MOT in satellite videos. From a unified reliability-aware spatiotemporal modeling perspective, RAST is designed to jointly mitigate temporal representation drift under degraded observations and reduce the task interference between detection and displacement estimation, thereby improving the robustness of subsequent data association.

Under this RAST perspective, satellite video tracking exhibits coupled challenges at multiple levels. *At the representation level*, tiny objects and complex backgrounds often lead to weak and unstable per-frame features, and naive one-way temporal modeling may accumulate historical noise, causing temporal drift. *At the optimization level*, detection and displacement estimation impose heterogeneous and even conflicting requirements on the shared feature space, resulting in a suboptimal tradeoff if treated with a single generic representation. *At the inference level*, the motion cue itself becomes reliability-variant: data-driven displacement regression may be noisy under degraded observations, while purely physics-driven prediction may lag

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

behind under nonlinear motion, making robust data association particularly challenging.

As illustrated in Fig. 2, RAST follows a progressive three-stage pipeline for satellite video MOT. Given an input sequence of K consecutive frames $\mathcal{I}_t = \{I_{t-K+1}, \ldots, I_t\}$ (with I_t denoting the current frame and the preceding $K-1$ frames providing temporal context), the framework first establishes a robust feature foundation [see Fig. 2(a)]. Specifically, a DVTEM aggregates forward motion inertia and backward contextual verification to produce temporally stabilized representations, which are then injected into the backbone for framewise feature extraction. Based on the resulting feature sequence, a forward convolutional GRU (ConvGRU)-based feature-level dynamic evolution encoder is further applied to model temporal state transitions in the backbone feature space.

Based on the enhanced temporal representations, RAST further constructs motion-guided decoupled representations [see Fig. 2(b)]. We maintain a temporal feature bank $\mathcal{F}_t = \{F_{t-K+1}, \ldots, F_t\}$ and employ an adaptive gated differencing encoder to extract explicit motion priors that guide dynamic feature aggregation. Two task-specific query streams are then constructed, i.e., a detection query and a displacement query, each interacting with the feature bank through deformable attention. This design yields two complementary representations: an appearance-oriented feature for object detection and a motion-oriented feature for cross-frame displacement estimation. Finally, the network outputs center heatmaps, box sizes, and center offsets for detection, together with the 2-D cross-frame displacement for tracking.

In the inference stage, RAST performs hybrid-driven cascade data association [see Fig. 2(c)]. Detections are first stratified by confidence scores. High-confidence detections are primarily associated with the network-predicted displacement (Stage I association), while low-confidence detections fall back to Kalman-filter-based inertial prediction and IoU gating (Stage II association) to improve robustness under degraded observations. The associated tracks are then updated and managed through a track lifecycle management module, producing the final track set with consistent identities across frames. To ensure the online capability, RAST operates in a strictly causal manner. For each time step t, the dual-view enhancement and subsequent modules only utilize information within the current observation I_t and the historical buffer. Consequently, the system maintains a zero-frame algorithmic lag, directly outputting the tracking results for the current frame t rather than a delayed frame.

B. Dual-View Temporal Enhancement Mechanism

The first stage of RAST aims to establish a robust temporal feature foundation by suppressing representation drift caused by tiny targets and degraded observations in satellite videos. To this end, we propose a DVTEM, as shown in Fig. 3. Given an input sequence of K consecutive frames $\mathcal{I}_t = \{I_{t-K+1}, \ldots, I_t\}$, DVTEM explicitly models two complementary temporal views: a forward view to capture motion inertia through progressive state propagation and a backward view operating within the current causal window to provide contextual verification from the current observation to preceding frames, thereby reducing the accumulation of historical noise.

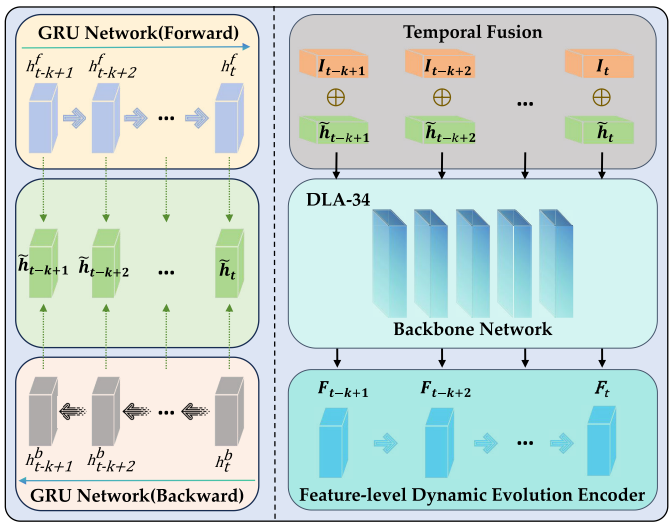


Fig. 3. Overview of the DVTEM. The left figure depicts the forward and backward ConvGRU branches with corrected temporal fusion; the right figure illustrates the subsequent pipeline comprising the backbone network and feature-level dynamic evolution encoder, yielding the temporal feature bank.

Specifically, both the forward and backward branches are implemented using a ConvGRU [39] encoder. The standard ConvGRU updates its hidden state $\mathbf{h}_\tau$ based on the input I_τ and the previous hidden state $\mathbf{h}_{\tau-1}$ as follows:

$$\begin{aligned}
\mathbf{z}_\tau &= \sigma\left(\mathbf{W}_z * I_\tau + \mathbf{U}_z * \mathbf{h}_{\tau-1}\right) \\
\mathbf{r}_\tau &= \sigma\left(\mathbf{W}_r * I_\tau + \mathbf{U}_r * \mathbf{h}_{\tau-1}\right) \\
\bar{\mathbf{h}}_\tau &= \tanh\left(\mathbf{W}_h * I_\tau + \mathbf{U}_h * (\mathbf{r}_\tau \odot \mathbf{h}_{\tau-1})\right) \\
\mathbf{h}_\tau &= (1 - \mathbf{z}_\tau) \odot \mathbf{h}_{\tau-1} + \mathbf{z}_\tau \odot \bar{\mathbf{h}}_\tau
\end{aligned} \tag{1}$$

where $*$ denotes the convolution, $\sigma(\cdot)$ is the sigmoid function, and $\odot$ is the elementwise product. The learnable kernels $\mathbf{W}_{\{\cdot\}}$ and $\mathbf{U}_{\{\cdot\}}$ map their respective inputs to the hidden dimension.

By applying this update recursively, the forward branch generates a sequence of hidden states $\{\mathbf{h}^f_{t-K+1}, \ldots, \mathbf{h}^f_t\}$, while the backward branch processes the input sequence in reverse to produce $\{\mathbf{h}^b_{t-K+1}, \ldots, \mathbf{h}^b_t\}$ after realigning to the original temporal order.

To couple these two views and suppress temporal drift, we perform a corrected temporal fusion for each time step τ

$$\tilde{\mathbf{h}}_\tau = \phi\left(\left[\mathbf{h}^f_\tau; \mathbf{h}^b_\tau\right]\right) \tag{2}$$

where $[\cdot;\cdot]$ denotes channelwise concatenation and the lightweight correction module $\phi(\cdot)$ is defined as

$$\phi(\mathbf{x}) = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}(\mathbf{x})\right)\right) \tag{3}$$

with $\mathrm{BN}(\cdot)$ representing batch normalization.

This fusion allows the backward view to provide an explicit contextual verification signal to rectify unreliable historical cues, while the forward view preserves motion inertia, resulting in temporally stabilized representations for subsequent feature extraction.

The corrected hidden states $\{\tilde{\mathbf{h}}_{t-K+1}, \ldots, \tilde{\mathbf{h}}_t\}$ produced by DVTEM serve as stabilized temporal context for the subsequent backbone feature extraction. Specifically, at each time

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

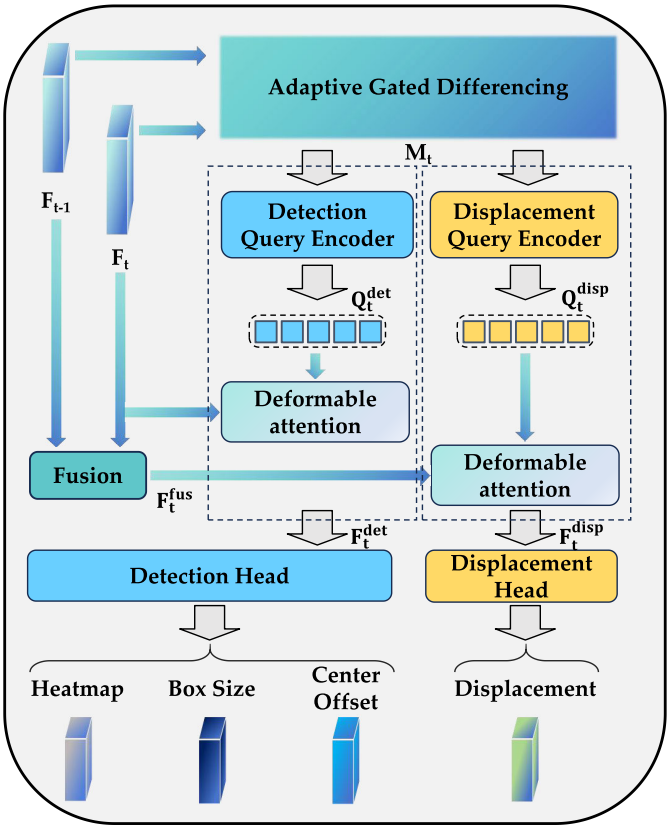


Fig. 4. MGDFAM: adaptive gated differencing extracts motion priors, and dual-query deformable attention decouples features for detection and displacement prediction.

step τ, $\tilde{\mathbf{h}}_\tau$ is elementwise fused with the corresponding raw frame I_τ to form a temporally conditioned input

$$\tilde{I}_\tau = I_\tau \oplus \mathrm{Proj}\left(\tilde{\mathbf{h}}_\tau\right) \tag{4}$$

where $\oplus$ denotes elementwise addition and Proj$(\cdot)$ is a lightweight projection layer that aligns the channel dimension of the corrected hidden state with that of the backbone input.

This fusion injects the rectified dual-view temporal context directly into the per-frame input stream, enabling the backbone to extract spatial features that are conditioned on both forward motion inertia and backward contextual verification, rather than relying solely on isolated frame appearance. The resulting enhanced sequence $\{\tilde{I}_{t-K+1}, \ldots, \tilde{I}_t\}$ is then processed frame-by-frame through the backbone network to produce per-frame spatial features. To further capture temporal state transitions across the backbone feature space, a forward ConvGRU-based feature-level dynamic evolution encoder is subsequently applied over the backbone output sequence, yielding the temporally evolved feature sequence $\{F_{t-K+1}, \ldots, F_t\}$. This sequence constitutes the temporal feature bank $\mathcal{F}_t = \{F_{t-K+1}, \ldots, F_t\}$, which serves as the input to the MGDFAM described in Section III-C.

C. Motion-Guided Dynamic Feature Aggregation Mechanism

The second stage of RAST aims to construct motion-guided decoupled representations for the heterogeneous tasks of object detection and cross-frame displacement estimation. As illustrated in Fig. 4, we maintain a short-term *temporal feature bank* $\mathcal{F}_t = \{F_{t-K+1}, \ldots, F_t\}$ that stores the enhanced feature states within the current causal window. Rather than performing dense global attention over the entire spatiotemporal volume, we leverage explicit motion priors to guide sparse and task-specific feature aggregation, which is particularly favorable for satellite videos with large fields of view and tiny targets.

From the temporal feature bank $\mathcal{F}_t$, which stores the feature states dynamically updated by the current DVTEM pass, we retrieve the two most recent feature maps, namely, $\mathbf{F}_t$ and $\mathbf{F}_{t-1}$. Crucially, unlike standard recursive updates where historical features remain static, our $\mathbf{F}_{t-1}$ is rectified by the dual-view enhancement, incorporating both historical inertia and current contextual feedback. This ensures that the computed absolute difference reflects reliable motion cues rather than temporal noise

$$\mathbf{D}_t = |\mathbf{F}_t - \mathbf{F}_{t-1}|. \tag{5}$$

To suppress background fluctuations and emphasize reliable motion-sensitive regions, we introduce an adaptive gating function that conditions on the local temporal context

$$\tilde{\mathbf{D}}_t = \sigma\left(\psi\left([\mathbf{F}_{t-1}; \mathbf{F}_t; \mathbf{D}_t]\right)\right) \odot \mathbf{D}_t \tag{6}$$

where $\psi(\cdot)$ is a lightweight bottleneck gating network, $\sigma(\cdot)$ is the sigmoid function, and $\odot$ denotes elementwise multiplication.

The gated difference $\tilde{\mathbf{D}}_t$ is then processed by a motion encoder to obtain a compact motion prior

$$\mathbf{M}_t = \gamma\left(\tilde{\mathbf{D}}_t\right) \tag{7}$$

where $\gamma(\cdot)$ is a shallow convolutional stack.

Based on the motion prior $\mathbf{M}_t$, we generate two task-specific queries

$$\mathbf{Q}_t^{det} = \eta_{det}\left(\mathbf{M}_t\right), \quad \mathbf{Q}_t^{disp} = \eta_{disp}\left(\mathbf{M}_t\right) \tag{8}$$

where $\eta_{det}(\cdot)$ and $\eta_{disp}(\cdot)$ are two independent query encoders.

We then perform deformable attention [40] in two decoupled branches.

Specifically, for a task-specific query $\mathbf{Q}_t$ at spatial location $\mathbf{p}$, the deformable attention aggregates features from M attention heads, each sampling K reference points

$$\mathrm{DeformAttn}\left(\mathbf{V}, \mathbf{Q}_t\right)(\mathbf{p}) = \sum_{m=1}^{M} \mathbf{W}_m \sum_{k=1}^{K} A_{mk} \cdot \mathbf{W}'_m \mathbf{V}\left(\mathbf{p} + \Delta\mathbf{p}_{mk}\right) \tag{9}$$

where $\Delta\mathbf{p}_{mk}$ are learned sampling offsets predicted from $\mathbf{Q}_t$. The attention weights A_{mk} are computed with temperature scaling

$$A_{mk} = \frac{\exp\left(\left(\mathbf{q}_m \cdot \mathbf{k}_{mk}\right)/\tau\right)}{\sum_{k'=1}^{K} \exp\left(\left(\mathbf{q}_m \cdot \mathbf{k}_{mk'}\right)/\tau\right)} \tag{10}$$

where τ is a temperature hyperparameter; dividing by $\tau < 1$ sharpens the softmax distribution, concentrating attention on fewer, more spatially precise locations—a design choice motivated by the need for accurate localization of tiny targets in cluttered satellite scenes.

Based on the above formulation, the two task-specific branches are instantiated as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

For detection, $\mathbf{F}_t$ serves as the value feature $\mathbf{V}$, and the motion-guided detection feature is obtained as

$$\mathbf{F}_t^{det} = \mathrm{Proj}_{det}(\mathbf{F}_t) + \mathrm{DeformAttn}\left(\mathbf{F}_t, \mathbf{Q}_t^{det}\right) \tag{11}$$

where DeformAttn$(\cdot)$ denotes deformable attention and $\mathrm{Proj}_{det}(\cdot)$ is a lightweight projection layer for stable optimization.

For displacement estimation, we first fuse the current and previous features to incorporate short-term temporal continuity

$$\mathbf{F}_t^{fus} = \rho([\mathbf{F}_t; \mathbf{F}_{t-1}]) \tag{12}$$

where $\rho(\cdot)$ is a small convolutional fusion module.

The displacement feature is then obtained by

$$\mathbf{F}_t^{disp} = \mathrm{Proj}_{disp}\left(\mathbf{F}_t^{fus}\right) + \mathrm{DeformAttn}\left(\mathbf{F}_t^{fus}, \mathbf{Q}_t^{disp}\right). \tag{13}$$

Finally, the detection head predicts center heatmaps, box sizes, and center offsets from $\mathbf{F}_t^{det}$, while the displacement head regresses a 2-D cross-frame displacement field from $\mathbf{F}_t^{disp}$. This motion-guided decoupling alleviates task interference and provides more reliable displacement cues for the subsequent hybrid-driven cascade association.

D. Hybrid-Driven Cascade Association Strategy

After obtaining high-quality detection and displacement features from the preceding stages, the final stage addresses the data association challenge under varying observation quality. The network-predicted displacement is highly adaptive to nonlinear motion for clear targets but can be noisy for low-confidence or occluded ones. Conversely, physics-based models, such as the SORT [21], offer stable predictions under degraded observations but often struggle to adapt to abrupt motion changes, such as stop-and-go maneuvers and sharp turns commonly seen in traffic scenarios. In practice, the detection confidence provides a convenient proxy for observation reliability, enabling us to make reliability-aware decisions without introducing extra estimation branches. To leverage the complementary strengths of both, we propose an HDCAS, summarized in Algorithm 1, which performs reliability-aware matching through confidence stratification and a two-stage association process.

The core mechanism consists of two key steps. First, *confidence stratification* partitions the detection set $\mathcal{D}_k$ into a high-confidence set $\mathcal{D}_h$ and a low-confidence set $\mathcal{D}_l$ based on score thresholds θ_h and θ_l. Second, a *two-stage matching* process is executed. Stage I performs displacement-guided association between the existing tracks $\mathcal{T}_k$ and high-confidence detections $\mathcal{D}_h$. The matching cost is primarily based on the proximity between the track's last observed position and the detection's position back-projected by the network-predicted displacement $\delta_d \in \Delta_k$. Stage II performs KF-guided association between the remaining unmatched tracks $\mathcal{T}_1^u$ and low-confidence detections $\mathcal{D}_l$. The cost is based on the Intersection over Union (IoU) between the KF-predicted bounding box $\hat{b}(\tau)$ and the detection box b_d.

In Stage I, the cost for an association pair (τ, d) is formulated as the Euclidean distance

$$d_{\mathrm{off}}(\tau, d) = \|(c_d - \delta_d) - c_\tau\|_2 \tag{14}$$

Algorithm 1 Hybrid-Driven Cascade Association Strategy

Input: Current-frame detections $\mathcal{D}_k = \{(b_i, s_i, cls_i)\}_{i=1}^{N_k}$, where b_i is the bounding box, s_i is the confidence score, and cls_i is the category label. Displacement predictions $\Delta_k = \{\delta_i\}_{i=1}^{N_k}$ aligned with detections in $\mathcal{D}_k$. Association thresholds θ_h (high) and θ_l (low). Track set from the previous frame $\mathcal{T}_{k-1}$, where each track $\tau \in \mathcal{T}_{k-1}$ stores (i) identity $id(\tau)$, (ii) latest observation $b(\tau), s(\tau), cls(\tau)$, and (iii) Kalman filter state $(\mu(\tau), \Sigma(\tau), \hat{b}(\tau))$.

Output: Updated track set $\mathcal{T}_k$ for the current frame k.

$\mathcal{T}_k \leftarrow \mathcal{T}_{k-1}$
// (a) Confidence-based partition
$\mathcal{D}_h \leftarrow \emptyset$, $\mathcal{D}_l \leftarrow \emptyset$
for $(b_i, s_i, cls_i) \in \mathcal{D}_k$ **do**
  if $s_i \geq \theta_h$ **then**
    $\mathcal{D}_h \leftarrow \mathcal{D}_h \cup \{(b_i, s_i, cls_i)\}$
  end
  else if $s_i \geq \theta_l$ **then**
    $\mathcal{D}_l \leftarrow \mathcal{D}_l \cup \{(b_i, s_i, cls_i)\}$
  end
end
// (b) KF prediction on $\mathcal{T}_k$
for $\tau \in \mathcal{T}_k$ **do**
  $(\mu(\tau), \Sigma(\tau)) \leftarrow \mathrm{KalmanPredict}(\mu(\tau), \Sigma(\tau))$
  $\hat{b}(\tau) \leftarrow \mathrm{BoxFromState}(\mu(\tau))$
end
// (c) Stage-1: High-confidence association (Displacement-guided)
$\mathcal{M}_1, \mathcal{T}_1^u, \mathcal{D}_h^u \leftarrow \mathrm{Associate}_{offset}(\mathcal{T}_k, \mathcal{D}_h, \Delta_k)$
for $(\tau, d) \in \mathcal{M}_1$ **do**
  $\mathrm{UpdateTrack}(\tau, d)$
end
// (d) Stage-2: Low-confidence association (KF-guided)
$\mathcal{M}_2, \mathcal{T}_2^u, \mathcal{D}_l^u \leftarrow \mathrm{Associate}_{iou}(\mathcal{T}_1^u, \mathcal{D}_l)$
for $(\tau, d) \in \mathcal{M}_2$ **do**
  $\mathrm{UpdateTrack}(\tau, d)$
end
// (e) Track management
$\mathcal{T}_k \leftarrow \mathrm{ManageTracks}(\mathcal{T}_k, \mathcal{T}_2^u)$
// (f) Initialize new tracks
foreach $d \in \mathcal{D}_h^u$ **do**
  $id \leftarrow id + 1$
  $\mathcal{T}_k \leftarrow \mathcal{T}_k \cup \{\mathrm{InitTrack}(d, id)\}$
end
return $\mathcal{T}_k$

where c_d and c_τ are the center coordinates of the detection and the track's last observation, respectively.

In Stage II, the cost is the IoU distance

$$d_{\mathrm{iou}}(\tau, d) = 1 - \mathrm{IoU}\left(\hat{b}(\tau), b_d\right). \tag{15}$$

In both stages, the assignment problem is solved using the Hungarian algorithm, with gating mechanisms (e.g., class consistency and distance/IoU thresholds) applied to prune infeasible matches. Unmatched tracks are managed by an

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

age-based lifecycle policy, while new tracks are initiated from all high-confidence detections that remain unmatched after the association stages.

This hybrid-driven cascade strategy provides two main benefits. First, it achieves robust motion modeling by adaptively switching between data- and physics-driven predictions without introducing extra network inference. Second, it enhances fault tolerance by using the KF as a safety net to recall targets with degraded appearance but consistent motion, thereby improving track continuity and reducing identity switches. This stage forms a closed loop with the preceding modules, effectively translating the high-quality representations into reliable and consistent tracklets. We note that unmatched high-confidence detections from Stage I are intentionally not routed to Stage II for a second matching attempt. This design choice is motivated by three satellite-specific considerations. First, under our reliability-aware principle, a high-confidence detection that fails to match via the network-predicted displacement—the most accurate motion prior available in our system—is more likely to represent a new instance entering the scene than an existing track and should, therefore, be initialized as a new trajectory. Second, Stage II relies on a linear Kalman filter, which represents a strictly weaker and less adaptive motion model than the learned displacement. Routing high-confidence detections to a weaker matching stage after they have already failed under the stronger prior is unlikely to produce reliable associations. Third, for the few-pixel targets in satellite videos, IoU-based matching is inherently brittle: a spatial offset of merely 2–3 pixels can reduce IoU to near zero, making Stage II unsuitable as a fallback for high-reliability detections in crowded scenes. This strict decoupling, therefore, prioritizes association precision and identity consistency under the reliability-variant conditions of satellite video MOT.

E. Loss Function

To enable effective multitask learning for satellite video tracking, RAST is trained with a weighted combination of a center heatmap classification term and several regression terms. Following the CenterNet [41] and CornerNet [42] practice, the center heatmap is supervised by a modified focal loss. Denoting the sigmoid-normalized predicted heatmap as $\hat{\mathbf{H}} \in [0,1]^{C\times H\times W}$ and the Gaussian target heatmap as $\mathbf{H}$, where $\hat{\mathbf{H}}_{cxy}$ and $\mathbf{H}_{cxy}$ denote the heatmap values at spatial location (x, y) of category c, the heatmap loss is formulated as

$$\mathcal{L}_{hm} = -\frac{1}{N_{pos}} \sum_{x,y,c} \Big(\mathbb{1}\left[\mathbf{H}_{cxy} = 1\right] \left(1 - \hat{\mathbf{H}}_{cxy}\right)^2 \log\left(\hat{\mathbf{H}}_{cxy}\right) + \mathbb{1}\left[\mathbf{H}_{cxy} < 1\right] \left(1 - \mathbf{H}_{cxy}\right)^4 \left(\hat{\mathbf{H}}_{cxy}\right)^2 \log\left(1 - \hat{\mathbf{H}}_{cxy}\right) \Big) \tag{16}$$

where N_{pos} denotes the number of ground-truth object centers (positive locations) on the heatmap and $(1 - \mathbf{H}_{cxy})^4$ down-weights easy negatives according to the Gaussian target value.

For the regression branches, supervision is applied only at ground-truth object centers. Specifically, $\mathcal{I} = \{i\}_{i=1}^{N}$ indexes the object centers on the feature map, and $\mathbf{m} \in \{0,1\}^N$ masks valid objects. Given a regression head $\hat{\mathbf{Y}}$ (e.g., size, offset, or displacement) and its target $\mathbf{Y}$, we gather predictions at $\mathcal{I}$ and use a masked ℓ_1 loss

$$\mathcal{L}_{reg}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) = \frac{1}{\sum_i \mathbf{m}_i + \epsilon} \sum_{i=1}^{N} \mathbf{m}_i \left\| \hat{\mathbf{Y}}_{\mathcal{I}_i} - \mathbf{Y}_i \right\|_1 \tag{17}$$

where ϵ is a small constant for numerical stability.

Specifically, the size head outputs $\hat{\mathbf{W}} \in \mathbb{R}^{2\times H\times W}$ representing object width and height, the offset head outputs $\hat{\mathbf{O}} \in \mathbb{R}^{2\times H\times W}$ to recover the discretization error, and the displacement head outputs $\hat{\mathbf{\Delta}} \in \mathbb{R}^{2\times H\times W}$ representing 2-D interframe offsets. Their corresponding ground-truth targets are denoted as $\mathbf{W}$, $\mathbf{O}$, and $\mathbf{\Delta}$, respectively. Accordingly, the size, offset, and displacement losses are defined as

$$\begin{aligned} \mathcal{L}_{wh} &= \mathcal{L}_{reg}\left(\hat{\mathbf{W}}, \mathbf{W}\right) \\ \mathcal{L}_{off} &= \mathcal{L}_{reg}\left(\hat{\mathbf{O}}, \mathbf{O}\right) \\ \mathcal{L}_{dis} &= \mathcal{L}_{reg}\left(\hat{\mathbf{\Delta}}, \mathbf{\Delta}\right). \end{aligned} \tag{18}$$

The overall objective is a weighted sum

$$\mathcal{L} = \lambda_{hm}\mathcal{L}_{hm} + \lambda_{wh}\mathcal{L}_{wh} + \lambda_{off}\mathcal{L}_{off} + \lambda_{dis}\mathcal{L}_{dis}. \tag{19}$$

IV. Experiments

A. Datasets and Metrics

To evaluate the proposed RAST under representative satellite video MOT scenarios, we conduct experiments on two widely adopted benchmarks: *SatVideoDT* [23] and *SatMTB-MOT* [24]. These datasets are collected from real satellite platforms and exhibit typical challenges in remote sensing videos, such as extremely small objects, complex backgrounds, subtle cross-frame displacement, and frequent occlusions. Therefore, they provide a practical benchmark for assessing both detection quality and long-term identity association ability.

For fair and reproducible comparison, we follow the official data protocols of each benchmark, including their standard splits and annotation formats. All methods are trained using the corresponding training sets and evaluated on the designated validation/test sets, ensuring that performance differences reflect model capability. Following the standard protocol in satellite MOT research, both benchmarks provide precropped video sequences (with resolutions primarily ranging from 512×512 to 1920×1080 pixels) rather than raw full-swath imagery.

The SatVideoDT dataset is a benchmark designed for moving object detection and tracking in satellite videos. It contains 114 video sequences with a total of 36 079 frames, captured by Jilin-1 satellite platforms. The benchmark is split into 82 training videos, 11 validation videos, and 21 test videos, where only the training and validation sets provide bounding-box annotations with instance identities. Since test annotations are not publicly accessible, we report results on the validation set following common practice. This dataset mainly focuses on vehicle targets and features dense target distributions, as well as fine-grained motion, which makes it suitable for analyzing tracking robustness under crowded small-object conditions.

The SatMTB-MOT dataset is a large-scale multiclass satellite video MOT benchmark reorganized from SatSOT [43]. It consists of 224 video sequences collected from multiple

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

satellite platforms, including Jilin-1, Skybox, and Carbonite-2. The dataset covers three object categories with diverse scales and appearances, including 92 vehicle videos, 70 ship videos, and 62 airplane videos. The official split follows a 6:4 ratio for training and testing. Compared with the SatVideoDT dataset, the SatMTB-MOT dataset provides richer object categories and broader scene variations, enabling a more comprehensive evaluation of multiclass and multiscale tracking performance.

We adopt standard MOT metrics [49] to quantitatively evaluate tracking quality from complementary perspectives. In particular, we report MOTA and IDF1 as the primary indicators, together with precision (Prcn), recall (Rcll), identity switches (IDs), mostly tracked (MT), and ML.

MOTA measures the overall tracking accuracy by jointly considering false negatives, false positives, and identity switches

$$\mathrm{MOTA} = 1 - \frac{\mathrm{FN} + \mathrm{FP} + \mathrm{IDs}}{\mathrm{GT}} \tag{20}$$

where FP, FN, IDs, and GT denote the number of false positives, false negatives, identity switches, and ground-truth objects, respectively. A higher MOTA indicates better overall performance, but it is relatively more sensitive to detection errors.

IDF1 focuses on identity consistency across time and evaluates association quality

$$\mathrm{IDF1} = \frac{2 \cdot \mathrm{IDTP}}{2 \cdot \mathrm{IDTP} + \mathrm{IDFP} + \mathrm{IDFN}} \tag{21}$$

where IDTP, IDFP, and IDFN represent identity-level true positives, false positives, and false negatives. In practice, IDF1 complements MOTA by emphasizing whether objects are consistently assigned correct identities over long sequences.

In addition, Prcn and Rcll reflect localization and detection reliability, and IDs directly indicate the frequency of identity mismatches, while MT and ML measure the number of trajectories that are tracked for more than 80% and less than 20% of their lifespan, respectively. Together, these metrics provide a comprehensive view of tracking accuracy, completeness, and identity preservation.

B. Implementation Details

RAST is implemented in PyTorch and uses a DLA-34 [50] based backbone. All experiments are conducted on two RTX 4090 GPUs. The input clip length is fixed to $k = 5$, which is commonly adopted in satellite video tracking (e.g., FMA-Net [29] and MP^{2}Net [11]). We train the network using AdamW with a batch size of 4 and an initial learning rate of 1.25×10^{-4}. The model is trained for a total of 20 epochs on the SatVideoDT dataset and 25 epochs on the SatMTB-MOT dataset. For both datasets, the learning rate is decayed by a factor of 0.1 at epochs 14 and 19. Sequence-consistent augmentation is applied, including random cropping to 512×512 and random horizontal/vertical flipping. For the loss function, the weighting factors in (19) are empirically set to $\lambda_{hm} = 1$, $\lambda_{wh} = 0.1$, $\lambda_{off} = 1$, and $\lambda_{dis} = 1$. The DLA-34 backbone is initialized with ImageNet-pretrained weights, while all other proposed components are trained from scratch without any domain-specific pretraining. During inference, we keep the original image resolution without test-time augmentation. For the cascade association, we set $\theta_h = 0.35$ and $\theta_l = 0.2$. The deformable attention is instantiated with $M = 4$ heads and $K = 4$ sampling points per head, with temperature scaling $\tau = 0.8$.

C. Comparison Results

For a fair and reproducible evaluation, all compared methods are trained and evaluated exclusively on the same satellite video benchmarks (SatVideoDT and SatMTB-MOT) under identical protocols. This ensures a rigorous and unbiased comparison across different approaches. We consider two types of MOT methods: generic methods developed for natural-scene benchmarks (DeepSORT [20], Tracktor [44], CenterTrack [16], FairMOT [45], and ByteTrack [22]) and satellite-oriented methods tailored for remote sensing videos (CFTracker [33], CKDNet-SMTNet [46], DSFNet [47], Hieum [48], MP^{2}Net [11], FMA-Net [29], and SPAX [14]). To preserve spatial details for small targets, we use an output stride of 2 whenever this setting is available in the original implementations. For DSFNet and Hieum, we follow their released inference pipelines and employ SORT [21] for online data association to obtain final trajectories. All remaining methods are evaluated with the default settings reported in their corresponding articles.

Table I reports the quantitative comparison on the SatVideoDT dataset. RAST achieves the best overall tracking accuracy with an MOTA of 67.0%. It also obtains the highest Rcll (75.1%) and the most MT (884) while yielding the lowest ML (137). Taken together, these results indicate that RAST maintains more complete and continuous trajectories in crowded satellite videos with tiny objects and cluttered backgrounds.

Overall, generic MOT methods developed for natural videos generally underperform satellite-oriented trackers on the SatVideoDT dataset, highlighting the domain gap caused by weak appearance cues, cluttered backgrounds, and dense interactions. Among satellite-oriented methods, RAST further shows clear advantages over the strongest baseline (SPAX [14]) on coverage-related metrics (Rcll/MT/ML). Specifically, it improves MOTA by 1.2 percentage points (67.0% versus 65.8%), boosts Rcll by 5.1 percentage points (75.1% versus 70.0%), increases MT by 100 (884 versus 784), and reduces ML by 35 (from 172 to 137). These gains directly reflect improved trajectory coverage and continuity, which are crucial on the SatVideoDT dataset where weak appearance cues and dense interactions frequently lead to missed detections and fragmented trajectories.

From a design perspective, the above trends are consistent with the goals of our RAST. The dual-view temporal enhancement is designed to stabilize temporal representations under degraded observations, which likely contributes to improved long-term continuity (higher MT) and reduced track loss (lower ML). Moreover, the motion-guided decoupled representation learning, together with the HDCAS, may help alleviate task interference and improve online matching robustness under crowded and noisy conditions, resulting in a favorable Prcn–Rcll balance (90.5% Prcn with the best Rcll of 75.1%). Overall, these compo-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE COMPARISON WITH SOTA METHODS ON THE SATVIDEODT DATASET. THE BEST AND SECOND BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Scene	Method	MOTA↑	IDF1↑	Rcll↑	Prcn↑	FP↓	FN↓	MT↑	ML↓	IDs↓
Natural	DeepSORT [20]	14.4%	32.2%	27.4%	71.0%	31448	204177	170	706	5048
	Tracktor [44]	6.0%	18.2%	17.2%	63.2%	28094	232901	43	927	3428
	FairMOT [45]	11.2%	21.3%	17.9%	82.9%	10399	230767	86	890	8423
	ByteTrack [22]	28.2%	45.6%	33.2%	87.1%	13782	187730	288	698	281
	CenterTrack [16]	38.3%	50.1%	57.5%	75.8%	51488	119617	601	311	2245
Remote Sensing	CFTracker [33]	35.4%	49.0%	59.3%	72.0%	64896	114434	621	292	2346
	CKDNet-SMTNet [46]	39.6%	57.7%	50.2%	83.1%	28759	140092	324	301	934
	DSFNet [47]	52.3%	66.1%	56.2%	93.8%	10418	123231	525	291	543
	Hieum [48]	59.5%	71.0%	62.6%	**95.6%**	**8089**	105241	619	207	490
	MP^{2}Net [11]	60.3%	72.0%	67.1%	91.1%	18330	92628	736	184	530
	FMA-Net [29]	65.5%	74.6%	73.2%	90.9%	20599	75257	854	139	1021
	SPAX [14]	65.8%	**76.3%**	70.0%	94.5%	11392	84470	784	172	**259**
	RAST(Ours)	**67.0%**	74.7%	**75.1%**	90.5%	22254	**69887**	**884**	**137**	637

TABLE II

QUANTITATIVE COMPARISON WITH SOTA METHODS ON THE SATMTB-MOT (VEHICLE) DATASET. THE BEST AND SECOND BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Scene	Method	MOTA↑	IDF1↑	Rcll↑	Prcn↑	FP↓	FN↓	MT↑	ML↓	IDs↓
Natural	DeepSORT [20]	11.2%	33.1%	27.9%	65.4%	50633	248045	335	1900	6822
	Tracktor [44]	15.9%	41.5%	34.9%	65.4%	63596	223843	500	1570	1626
	CenterTrack [16]	35.7%	45.2%	38.3%	**95.3%**	**6479**	212170	760	1580	2438
Remote Sensing	CFTracker [33]	48.0%	60.3%	63.4%	81.1%	50854	125719	1546	751	2325
	CKDNet-SMTNet [46]	25.6%	46.7%	37.9%	76.2%	40776	213647	242	1136	1352
	DSFNet [47]	55.9%	68.9%	59.6%	94.6%	11817	138889	1295	790	939
	Hieum [48]	56.1%	70.7%	**74.0%**	81.0%	59575	**89277**	**1976**	**407**	2198
	MP^{2}Net [11]	55.8%	68.7%	59.3%	94.7%	11355	139824	1344	878	739
	FMA-Net [29]	59.9%	71.7%	67.3%	90.9%	23192	112584	1699	581	2078
	SPAX [14]	57.2%	70.8%	63.3%	91.3%	20663	126066	1449	726	**577**
	RAST(Ours)	**63.0%**	**71.9%**	69.8%	91.7%	21783	103832	1768	504	1653

nents jointly enhance the robustness of the tracking system under challenging conditions.

Regarding identity consistency, RAST achieves an IDF1 of 74.7%, slightly below the best result (SPAX [14]: 76.3%). This gap is reasonable when interpreted together with complementary indicators. The SatVideoDT dataset contains frequent close-proximity interactions among tiny objects, where identity assignment becomes particularly sensitive to short-term fragmentation, while MOTA and coverage-related metrics (Rcll/MT/ML) better reflect the ability to continuously recover targets. Notably, compared with another confidence-stratified two-stage satellite tracker, FMA-Net [29], RAST yields fewer identity switches and a marginally higher IDF1 (IDs: 637 versus 1021; IDF1: 74.7% versus 74.6%). It also achieves higher MOTA and Rcll (MOTA: 67.0% versus 65.5%; Rcll: 75.1% versus 73.2%).

Fig. 5 visualizes tracking results on the SatVideoDT dataset across two key frames (100th and 300th) and their corresponding trajectories. To further illustrate the above quantitative trends, the zoomed-in regions focus on road intersections, which are typical yet challenging areas in satellite videos due to dense traffic flow and frequent crossing/merging motions, making reliable association challenging for tiny objects. As highlighted in the zoomed-in regions, RAST produces fewer missed targets (yellow boxes) while maintaining a low level of false alarms (red boxes) compared with other satellite-oriented trackers, indicating a better balance between target coverage and false-positive suppression in crowded tiny-object scenes. Furthermore, the trajectory visualization demonstrates that RAST maintains more continuous tracklets with fewer long-term breaks through dense interactions across time, qualitatively supporting the improvements in Rcll, MT, and ML

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.



Fig. 5. Qualitative comparison of tracking results on SatVideoDT (video 002). Columns correspond to MP^{2}Net, FMA-Net, SPAX, and ours (from left to right). From top to bottom, results are visualized at the 100th frame, the 300th frame, and the trajectory visualization. The highlighted regions are shown in zoomed-in views for better inspection. Green boxes indicate TP, red boxes denote FP, and yellow boxes represent FN.

on the SatVideoDT dataset. These observations align with the design of RAST, where dual-view temporal enhancement mitigates representation drift across frames, and the HDCAS handles ambiguous observations through adaptive matching strategies.

To further evaluate cross-category generalization under multiclass and multiscale variations, we conduct experiments on the SatMTB-MOT dataset. As shown in Tables II–IV, RAST achieves the best MOTA on all three subsets. It also maintains strong identity performance across categories, achieving the best IDF1 on vehicle and airplane and the second-best IDF1 on ship. These consistent advantages on the primary metrics indicate that RAST adapts well to category-dependent appearance and motion differences, demonstrating robust generalization across different target categories.

On the vehicle subset (see Table II), RAST achieves the best MOTA (63.0%) and the best IDF1 (71.9%), indicating strong overall tracking accuracy and identity consistency

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE COMPARISON WITH SOTA METHODS ON THE SATMTB-MOT (SHIP) DATASET. THE BEST AND SECOND BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Scene	Method	MOTA↑	IDF1↑	Rcll↑	Prcn↑	FP↓	FN↓	MT↑	ML↓	IDs↓
Natural	DeepSORT [20]	41.8%	65.7%	63.5%	74.7%	4687	7391	45	17	40
	Tracktor [44]	30.5%	55.3%	63.9%	66.6%	6975	7855	38	15	280
	CenterTrack [16]	42.1%	63.6%	55.8%	80.4%	2959	9599	36	22	20
Remote Sensing	CFTracker [33]	29.6%	63.9%	63.9%	65.1%	7436	7856	48	22	**5**
	CKDNet-SMTNet [46]	17.1%	39.3%	29.5%	70.8%	2647	15335	15	38	42
	DSFNet [47]	41.5%	60.3%	53.6%	82.2%	2516	10089	28	27	108
	Hieum [48]	24.3%	47.2%	37.0%	74.7%	2725	13706	22	40	24
	MP^{2}Net [11]	51.1%	64.9%	62.7%	**84.8%**	**2433**	8113	44	18	82
	FMA-Net [29]	52.6%	70.7%	**73.2%**	78.3%	4408	**5824**	**54**	**9**	76
	SPAX [14]	51.8%	**72.5%**	68.4%	80.5%	3600	6861	49	16	10
	RAST(Ours)	**56.1%**	71.2%	71.9%	82.2%	3374	6115	52	13	62

TABLE IV

QUANTITATIVE COMPARISON WITH SOTA METHODS ON THE SATMTB-MOT (AIRPLANE) DATASET. THE BEST AND SECOND BEST RESULTS ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Scene	Method	MOTA↑	IDF1↑	Rcll↑	Prcn↑	FP↓	FN↓	MT↑	ML↓	IDs↓
Natural	DeepSORT [20]	51.9%	74.2%	70.1%	79.4%	2837	4676	43	13	3
	Tracktor [44]	17.2%	57.0%	60.3%	58.4%	6701	6204	31	19	24
	CenterTrack [16]	49.8%	71.4%	63.1%	82.6%	2075	5770	40	21	2
Remote Sensing	CFTracker [33]	44.1%	71.7%	70.7%	72.7%	4152	4580	45	16	5
	CKDNet-SMTNet [46]	50.8%	66.4%	52.6%	**96.8%**	**271**	7396	33	17	14
	DSFNet [47]	63.3%	77.2%	66.9%	95.1%	544	5166	40	9	17
	Hieum [48]	53.9%	71.0%	59.3%	92.0%	808	6351	36	12	34
	MP^{2}Net [11]	71.6%	83.9%	77.6%	92.8%	937	3491	50	7	7
	FMA-Net [29]	73.3%	84.8%	83.2%	89.5%	1531	2620	53	**3**	19
	SPAX [14]	61.0%	81.4%	**85.4%**	77.7%	3818	**2277**	**55**	6	**0**
	RAST(Ours)	**73.4%**	**85.4%**	80.1%	92.4%	1027	3115	51	7	7

for dense tiny-vehicle tracking. Beyond the primary metrics, RAST also maintains strong trajectory coverage and continuity, as reflected by Rcll (69.8%), a low number of false negatives (FN = 1 03 832), and stable long-term statistics (MT = 1768 and ML = 504). These indicators are particularly important for satellite vehicle tracking, where frequent close-proximity interactions and background clutter easily cause missed detections and fragmented tracks. These observations are in line with the proposed design, where dual-view temporal enhancement and HDCAS improve robustness under degraded observations, and motion-guided decoupling helps mitigate task interference between detection and displacement estimation. Fig. 6 further visualizes trajectory comparisons on two vehicle sequences, where the zoomed-in regions focus on road junctions with frequent crossing and merging motions. In these interaction-intensive regions with frequent direction changes, RAST exhibits more continuous tracklets than the other compared methods in the zoomed-in views. This visual evidence qualitatively corroborates the strong coverage and continuity statistics reported in Table II (Rcll/FN and MT/ML).

On the ship subset (see Table III), RAST achieves the best MOTA of 56.1% and maintains strong identity consistency with an IDF1 of 71.2%. Compared with vehicles, ship targets are generally larger, yet maritime scenes remain challenging due to cluttered backgrounds, low-contrast appearances, and subtle cross-frame displacement, all of which can degrade observations and complicate association. In this setting, RAST demonstrates robust detection quality and trajectory coverage, as reflected by Rcll (71.9%), Prcn (82.2%), and a low number of false negatives (FN = 6115), together with favorable trajectory-level statistics (MT = 52 and ML = 13). We note that identity-related indicators may exhibit different strengths across methods on this subset (e.g., SPAX [14] achieves the best IDF1, while CFTracker [33] reports the lowest IDs), reflecting the inherent difficulty of identity preservation under subtle motion and ambiguous visual cues. Fig. 7 further

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

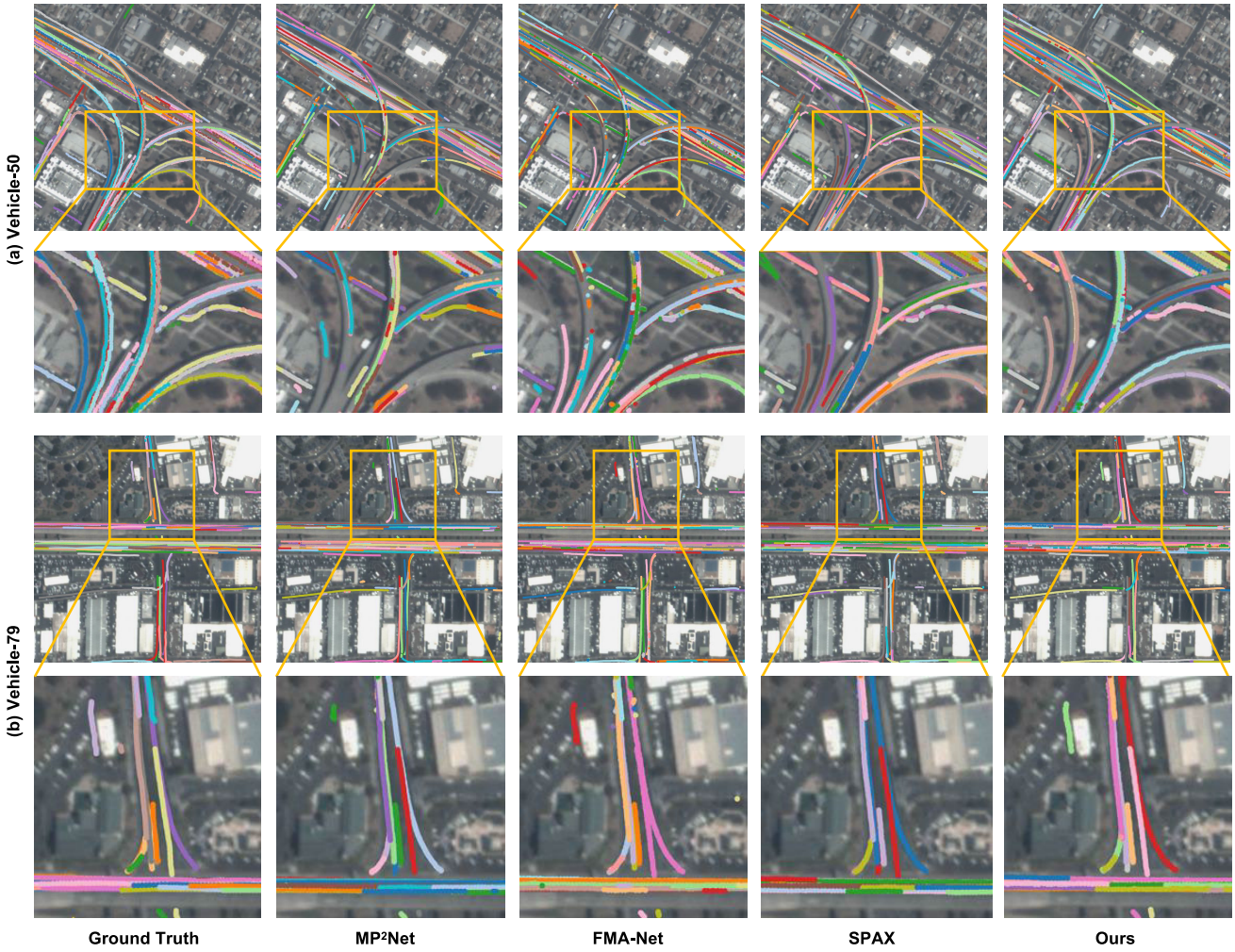


Fig. 6. Qualitative trajectory comparison on SatMTB-MOT (Vehicle). Two representative sequences [(a) Vehicle-50 and (b) Vehicle-79] are shown. From left to right: ground truth, MP^{2}Net, FMA-Net, SPAX, and our RAST. For each sequence, the highlighted region is further zoomed in for better inspection.

provides qualitative comparisons on two ship sequences (Ship-03 and Ship-57). In Ship-03, the compared methods tend to suffer from noticeable localization deviations, which may lead to both false positives and false negatives under IoU-based evaluation; in contrast, RAST provides more accurate localization. In the more challenging Ship-57 scene with multiple small targets and a cluttered background, RAST maintains better target coverage with fewer missed ships. These observations qualitatively corroborate the strong Rcll/Prcn and low FN reported in Table III, as well as the best MOTA achieved on the ship subset.

On the airplane subset (see Table IV), RAST achieves the best MOTA (73.4%) and the best IDF1 (85.4%). Compared with vehicles and ships, airplane targets are generally larger and more distinguishable, and the number of simultaneously moving instances is typically smaller. As a result, the qualitative differences among different trackers can be less pronounced in some sequences. Nevertheless, as illustrated in Fig. 8, RAST provides stable tracking results across frames, supporting its top performance on the primary accuracy and identity metrics reported in Table IV.

Overall, the comprehensive experiments on the SatVideoDT dataset and the SatMTB-MOT dataset consistently demonstrate the effectiveness and robustness of RAST for satellite video MOT. On the SatVideoDT dataset, RAST achieves strong tracking accuracy and maintains better trajectory coverage and continuity under crowded tiny-object scenarios, as supported by both quantitative results (see Table I) and qualitative visualization (Fig. 5). On the more diverse SatMTB-MOT dataset with pronounced multiclass and multiscale variations, RAST demonstrates robust cross-category generalization, achieving the best MOTA across vehicle, ship, and airplane subsets and leading identity performance. The corresponding qualitative results (see Figs. 6–8) further visually support these quantitative conclusions by showing more stable and continuous tracklets under complex interactions and degraded observations. Moreover, Fig. 9 provides a multiclass MOT visualization across three categories, further illustrating the robustness of RAST under diverse satellite video scenarios.

In addition to tracking accuracy, we further compare the computational efficiency of representative satellite-oriented trackers on SatVideoDT in terms of parameter count, FLOPs,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

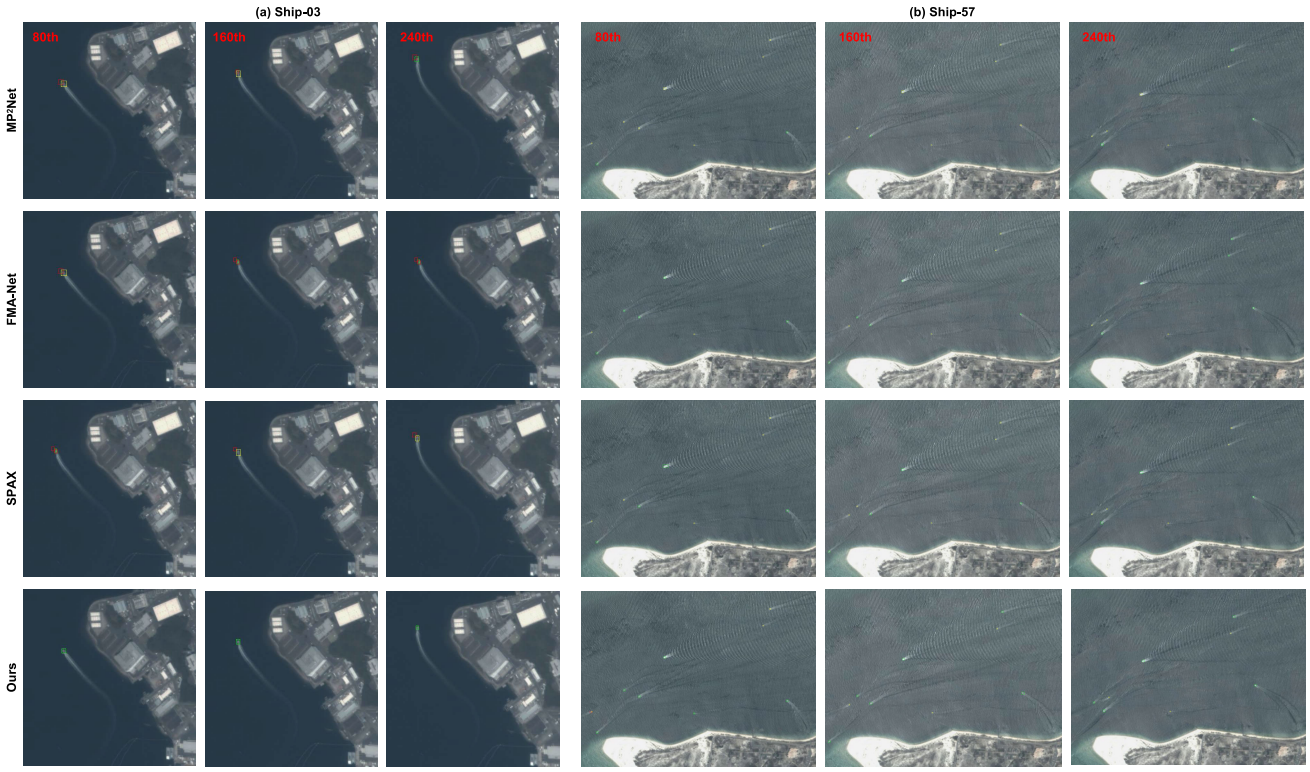


Fig. 7. Qualitative comparison of tracking results on two representative sequences from SatMTB-MOT (Ship) (a) Ship-03, a scene with a single prominent target, and (b) Ship-57, a more cluttered scene with multiple small targets. (From top to bottom) MP^{2}Net, FMA-Net, SPAX, and our RAST. Green boxes denote TP, while red boxes denote FP, and yellow boxes indicate FN. Our method demonstrates more accurate localization in (a) and better target coverage in (b).



Fig. 8. Qualitative comparison of airplane tracking results on SatMTB-MOT. From top to bottom: MP^{2}Net, FMA-Net, SPAX, and our RAST. Green boxes denote TP, red boxes denote FP, and yellow boxes denote FN.

TABLE V

MODEL COMPLEXITY AND RUNTIME COMPARISON ON THE SATVIDEODT DATASET. FPS IS MEASURED ON AN RTX 4090 GPU UNDER THE SAME INFERENCE CONFIGURATION FOR ALL METHODS

Method	Params(M)↓	FLOPs(G)↓	FPS ↑
MP^{2}Net [11]	6.7	602	2.4
FMA-Net [29]	8.4	583	0.9
SPAX [14]	7.9	522	4.8
RAST(Ours)	6.9	528	1.5

and runtime, as summarized in Table V. RAST contains 6.9M parameters and requires 528 GFLOPs, which are comparable to SPAX [14] (7.9M parameters and 522 GFLOPs) and lower than FMA-Net [29] (8.4M parameters and 583 GFLOPs). Compared with MP^{2}Net [11], RAST uses a similar number of parameters (6.9 versus 6.7 M) with fewer FLOPs (528 versus 602 G). These results suggest that the performance improvements of RAST are primarily due to the proposed RAST design rather than increased model capacity. RAST runs at 1.5 FPS on an RTX 4090 GPU, which is slower than SPAX [14] (4.8 FPS) but faster than FMA-Net [29] (0.9 FPS). The lower runtime mainly comes from the sequential temporal processing in DVTEM and the feature-level dynamic evolution encoder, which operate causally to stabilize temporal representations under degraded satellite observations. In contrast, the fully sparse design of SPAX [14] is more favorable for inference throughput. Overall, RAST achieves the best tracking performance on SatVideoDT with competitive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

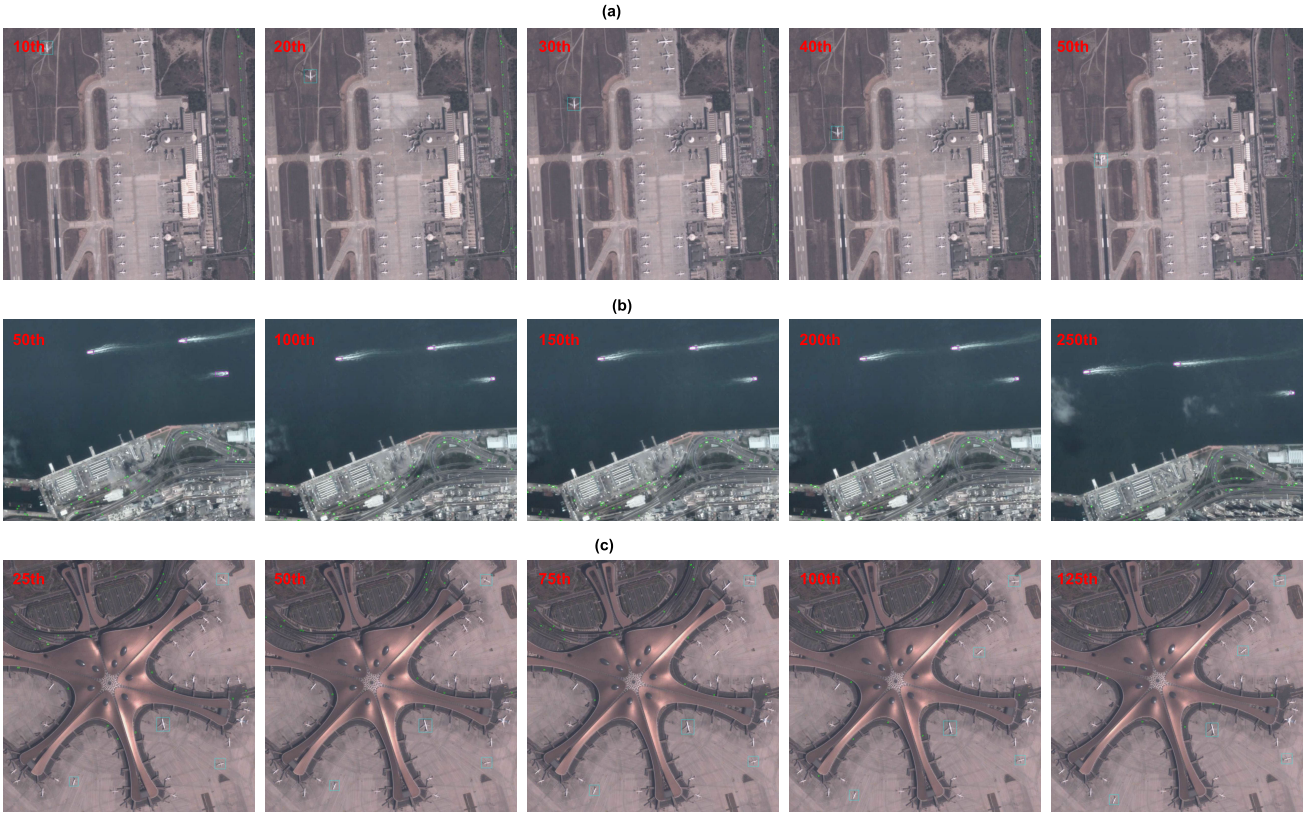


Fig. 9. Multiclass qualitative visualization of RAST on SatMTB-MOT. Three sequences are shown (a) Vehicle-03, (b) Ship-04, and (c) Airplane-46. Multiple frames are sampled from each sequence to illustrate the multiclass MOT performance of RAST.

model complexity while incurring additional runtime cost for more robust temporal modeling.

D. Ablation Studies

To validate our RAST framework, we conduct a series of ablation studies on the SatVideoDT dataset. We first dissect the framework by progressively integrating DVTEM, MGDFAM, and HDCAS into a CenterTrack-style baseline, quantifying their individual contributions and synergistic effects under crowded tiny-object tracking. We then perform in-depth analyses on the core mechanism of each key design: 1) for DVTEM, we investigate temporal modeling variants to verify the necessity of dual-view fusion; 2) for MGDFAM, we ablate the task-specific decoupled architecture and the gated motion-prior construction strategy to verify the necessity of motion-guided dynamic feature aggregation for mitigating task interference; and 3) for HDCAS, we isolate its contribution by comparing it against other association strategies using identical network predictions and further analyze the sensitivity of its reliability-aware confidence stratification. In addition, intermediate feature visualization is provided to offer intuitive evidence for temporal stabilization and the learned motion prior.

Table VI summarizes the overall contribution of each component and their combinations. Starting from the baseline (60.3% MOTA and 63.4% IDF1), integrating all components yields 67.0% MOTA and 74.7% IDF1, together with improved Rcll (75.1%), more MT (884), fewer ML (137), and substantially fewer identity switches (IDs = 637). These results demonstrate that RAST improves overall tracking accuracy, trajectory coverage, and long-term identity consistency under crowded tiny-object scenarios.

The improvements follow a clear componentwise pattern consistent with our design goals. Adding DVTEM alone brings a notable boost in MOTA (from 60.3% to 65.5%) and Prcn (from 87.7% to 91.0%), suggesting that dual-view temporal enhancement effectively stabilizes per-frame representations and suppresses drift under degraded observations. In contrast, enabling HDCAS alone leads to a much stronger improvement in identity-related metrics (IDF1: from 63.4% to 70.9% and IDs: from 2026 to 736), highlighting the importance of reliability-aware association that adaptively combines data-driven displacement and physics-driven inertial prediction. This effect is further isolated and verified in the association-strategy ablation (see Table IX). Notably, the combination of DVTEM and HDCAS already yields strong trajectory-level benefits (e.g., MT = 919 and ML = 123), indicating that stabilized temporal representations and reliability-aware association play complementary and mutually reinforcing roles in satellite video MOT. We further analyze the interaction between modules, which reveals a distinct hierarchical synergy. As shown in Table VI, enabling MGDFAM alone yields limited gains, suggesting that explicit motion modeling is sensitive to noise in raw feature representations. However, a remarkable synergy emerges when MGDFAM is paired with DVTEM. This combination achieves the highest Prcn

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION STUDY OF THE PROPOSED COMPONENTS IN RAST ON THE SATVIDEODT DATASET

Baseline	DVTEM	MGDFAM	HDCAS	MOTA↑	IDF1↑	Rcll↑	Prcn↑	MT↑	ML↓	IDs↓
✓				60.3%	63.4%	70.9%	87.7%	829	157	2026
✓	✓			65.5%	66.7%	73.4%	91.0%	860	140	1772
✓		✓		58.2%	62.1%	71.4%	85.2%	837	146	2120
✓			✓	60.1%	70.9%	74.2%	84.2%	888	139	736
✓	✓	✓		65.6%	65.8%	71.7%	92.9%	828	157	1939
✓	✓		✓	64.3%	74.1%	76.6%	86.4%	919	123	627
✓		✓	✓	58.2%	71.1%	74.7%	82.2%	888	125	764
✓	✓	✓	✓	67.0%	74.7%	75.1%	90.5%	884	137	637

TABLE VII
ABLATION STUDY OF TEMPORAL MODELING VARIANTS IN DVTEM ON THE SATVIDEODT DATASET, INCLUDING NONE, FORWARD-ONLY, BACKWARD-ONLY, AND DUAL-VIEW FUSION

Temporal	MOTA↑	IDF1 ↑	Rcll ↑	Prcn ↑	MT ↑	ML ↓	FLOPs(G) ↓	FPS ↑
None	58.2%	71.1%	74.7%	82.2%	888	125	-	-
Forward-only	66.5%	75.0%	73.3%	91.8%	863	142	447	1.6
Backward-only	61.5%	71.2%	74.3%	85.6%	864	131	447	1.6
Dual-view	67.0%	74.7%	75.1%	90.5%	884	137	528	1.5

(92.9%) among all variants, indicating that once temporal drift is mitigated by DVTEM, MGDFAM effectively exploits clean motion priors to suppress background false alarms. Although this high-Prcn setting becomes slightly conservative (resulting in lower Rcll), the inclusion of HDCAS unlocks the full potential of the framework. HDCAS compensates for the conservative detection by adaptively recovering hard targets through cascade association, boosting Rcll to 75.1%, and achieving the best overall MOTA (67.0%) and IDF1 (74.7%). This demonstrates that DVTEM establishes a stable foundation, MGDFAM purifies the representation, and HDCAS ensures robust recovery, working complementarily to address the challenges in satellite videos.

Fig. 10 provides qualitative support for the proposed RAST. The visualization suggests that the dual-view temporal enhancement mitigates representation drift and background interference under cluttered satellite scenes, making tiny targets more salient and temporally stable. Meanwhile, the motion prior in MGDFAM exhibits the cross-frame motion of targets, providing a complementary motion cue for motion-guided feature aggregation. Consequently, the decoded center heatmaps show clearer and more stable target responses. These observations are consistent with the quantitative gains in trajectory coverage and continuity reported in Table VI.

Table VII evaluates different temporal modeling variants in DVTEM. Without temporal enhancement, the tracker suffers from a much less favorable Prcn–Rcll tradeoff under cluttered satellite scenes, resulting in a low MOTA of 58.2% and a Prcn of 82.2%. Introducing a forward-only branch substantially improves MOTA to 66.5% and yields the highest Prcn (91.8%) together with the best IDF1 (75.0%), suggesting that causal state propagation effectively captures motion inertia and stabilizes short-term representations for confident detection and association. By contrast, the backward-only variant provides more limited gains (61.5% MOTA and 85.6% Prcn) and yields nearly unchanged identity quality (71.2% IDF1 versus 71.1% without temporal enhancement), implying that contextual verification alone is insufficient to maintain strong causal temporal evolution under dense interactions and subtle motion. Importantly, the proposed dual-view fusion achieves the best overall MOTA (67.0%) and the strongest Rcll (75.1%) while maintaining competitive identity quality (74.7% IDF1) and improved trajectory continuity (MT: from 863 to 884). Compared with forward-only propagation, the dual-view design recovers more hard targets over time, which is consistent with our motivation that backward contextual verification can rectify unreliable historical cues and suppress drift accumulation, especially when tiny targets are intermittently weak or partially occluded. Overall, the results suggest that forward propagation favors conservative and identity-stable tracking (higher Prcn/IDF1), while dual-view fusion better improves target recovery (higher Rcll and the best MOTA), validating the necessity of coupling forward motion inertia and backward contextual verification in DVTEM for robust satellite video MOT. Furthermore, we evaluate the cost–benefit tradeoff of the dual-view design in Table VII. Compared to the forward-only variant, the dual-view configuration introduces an additional 81G FLOPs (18.1% increase), while the actual inference speed decreases only marginally by 0.1 FPS (from 1.6 to 1.5). Importantly, this modest computational investment yields a qualitative improvement in tracking reliability, with the number of mostly tracked (MT) targets increasing by 21 (from 863 to 884). For high-precision satellite surveillance where maintaining long-term identity is paramount, the enhanced robustness provided by dual-view verification well justifies its additional computational cost.

Table VIII examines MGDFAM from two orthogonal design dimensions, i.e., task decoupling (shared versus decoupled) and motion-prior construction (hard difference versus soft gated difference). Overall, the results suggest that task decoupling is the primary factor for improving both tracking accuracy and identity consistency, while gated differencing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

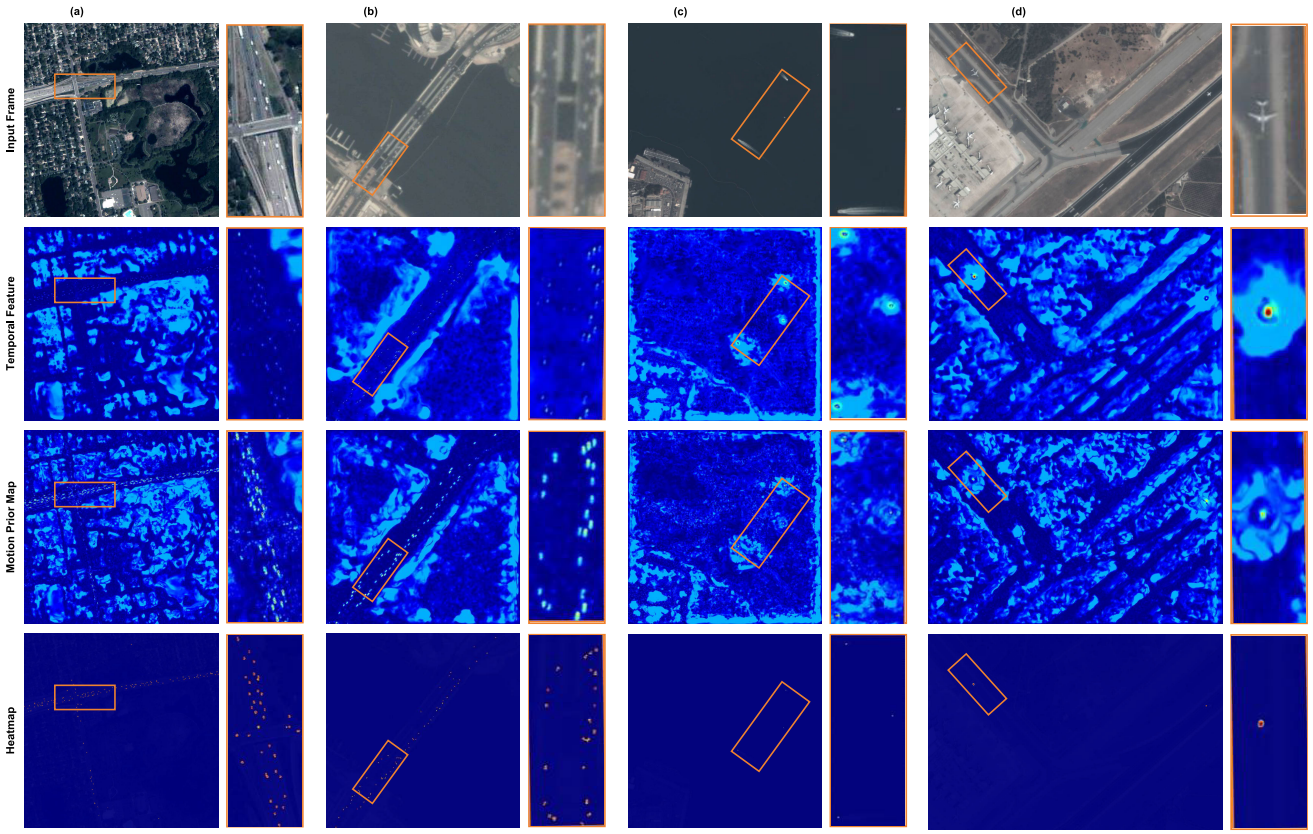


Fig. 10. Feature visualization of RAST on representative sequences from SatVideoDT and SatMTB-MOT. From top to bottom: the raw input frame; temporally enhanced features produced by the proposed DVTEM; the motion prior map in MGDFAM, generated by the gated difference and motion encoder, which exhibits the cross-frame motion of targets; and the decoded center heatmap, in which peaks indicate target center localization. Orange bounding boxes indicate the regions of interest (ROIs), with the corresponding enlarged views shown on the right of each map for better inspection of tiny targets. (a) SatVideoDT-010. (b) SatMTB-Vehicle-23. (c) SatMTB-Ship-50. (d) SatMTB-Airplane-15.

further refines the motion prior to yield cleaner detections. Regarding task decoupling, switching from shared to decoupled representations consistently brings large improvements under both differencing settings. With hard differencing, decoupling boosts MOTA from 59.6% to 66.3% and IDF1 from 70.0% to 75.2%, accompanied by a substantial reduction in false negatives (FN: from 91 854 to 68 541), indicating better target recovery and more stable identity assignment. Similar gains are observed with soft gated differencing, where decoupling improves MOTA (from 63.9% to 67.0%) and Prcn (from 87.7% to 90.5%). These trends are consistent with our motivation that detection and displacement estimation impose heterogeneous objectives and, thus, benefit from separating appearance- and motion-oriented representations in MGDFAM. Regarding motion-prior construction, replacing hard differencing with soft gated differencing yields a more reliable motion cue for aggregation. Under the decoupled setting, soft gating further improves MOTA (from 66.3% to 67.0%) and Prcn (from 89.2% to 90.5%) and reduces FP (from 25 678 to 22 254), suggesting that gating suppresses spurious motion responses induced by noisy cross-frame variations. We also observe a slight increase in FN (from 68 541 to 69 887), implying a Prcn–Rcll tradeoff where a cleaner and more selective motion prior reduces false alarms while being mildly more conservative on hard targets. Overall, combining decoupled representations with soft gated differencing achieves the best performance (67.0% MOTA and 74.7% IDF1), which is consistent with the overall ablation in Table VI and supports MGDFAM as an effective motion-guided decoupling module within RAST.

TABLE VIII

ABLATION STUDY OF MGDFAM ON THE SATVIDEODT DATASET WITH SHARED/DECOUPLED ARCHITECTURES AND HARD/SOFT-GATED DIFFERENCING

Architecture Type	Diff	MOTA↑	IDF1 ↑	Rcll ↑	Prcn ↑	FP↓	FN↓
Shared	Hard diff	59.6%	70.0%	67.3%	89.9%	21231	91854
Decoupled	Hard diff	66.3%	75.2%	75.6%	89.2%	25678	68541
Shared	Soft gated diff	63.9%	74.0%	74.6%	87.7%	29488	71372
Decoupled	Soft gated diff	67.0%	74.7%	75.1%	90.5%	22254	69887

Table IX compares different online association strategies under the same detection and displacement predictions to isolate the contribution of association design. As shown in the table, the proposed HDCAS achieves the best overall performance on the SatVideoDT dataset, obtaining the highest MOTA (67.0%) and IDF1 (74.7%), together with the best trajectory coverage and continuity (Rcll = 75.1%, MT = 884, and ML = 137). Even when some score-driven association baselines yield comparable MOTA, they tend to show weaker identity consistency and coverage statistics, indicating that robust association under reliability-variant observations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
COMPARISON OF DIFFERENT DATA ASSOCIATION STRATEGIES ON THE SATVIDEODT DATASET. DETECTION AND DISPLACEMENT PREDICTIONS ARE FIXED, AND ONLY THE ASSOCIATION STRATEGY IS CHANGED. THE BEST RESULTS ARE SHOWN IN BOLD

Association strategy	MOTA↑	IDF1 ↑	Rcll ↑	MT ↑	ML ↓	IDs ↓
SORT-style	64.6%	74.3%	67.0%	737	191	**542**
ByteTrack-style	66.4%	72.6%	72.2%	831	157	2169
CenterTrack-style	65.6%	65.8%	71.7%	828	157	1939
Ours (HDCAS)	**67.0%**	**74.7%**	**75.1%**	**884**	**137**	637

TABLE X
SENSITIVITY ANALYSIS OF CONFIDENCE THRESHOLDS (θ_h, θ_l) FOR RELIABILITY-AWARE STRATIFICATION IN HDCAS ON THE SATVIDEODT DATASET

θ_h	θ_l	MOTA ↑	IDF1 ↑	Rcll ↑	MT ↑	ML ↓	IDs ↓
0.45	0.2	64.7%	75.1%	67.8%	755	243	278
0.4	0.2	67.9%	76.3%	73.4%	847	157	405
0.35	0.2	67.0%	74.7%	75.1%	884	137	637
0.3	0.2	65.3%	71.6%	76.0%	899	129	1086
0.35	0.3	66.5%	71.3%	71.1%	810	162	1055
0.35	0.25	67.3%	74.1%	73.4%	852	152	758
0.35	0.15	64.8%	73.4%	76.7%	909	128	598

remains a key bottleneck in satellite video MOT. To address this bottleneck, HDCAS leverages confidence stratification to switch between two complementary matching cues. Specifically, for high-confidence detections, displacement-guided association is preferred. Unlike the inertial assumption of KF, the network-predicted displacement captures instantaneous motion intentions, allowing it to better adapt to complex nonlinear trajectories and sudden speed variations in crowded scenes. In contrast, for low-confidence detections, HDCAS falls back to Kalman-filter-based prediction with IoU gating as a safety net, improving fault tolerance when displacement estimates become unreliable. Consequently, HDCAS achieves a favorable balance between agility (accurate matching under clear observations) and robustness (stable recovery under degraded observations), effectively translating the stabilized representations learned by DVTEM and the motion-aware cues learned by MGDFAM into more reliable tracklet association.

Table X analyzes the sensitivity to the confidence thresholds (θ_h, θ_l) used for reliability stratification in HDCAS. Specifically, θ_h determines which detections are considered reliable and, thus, associated in Stage-I using displacement cues, while θ_l controls which low-confidence detections are still admitted to the Stage-II Kalman-based fallback. The results reveal an interpretable tradeoff between trajectory coverage and identity conservativeness: a higher θ_h (e.g., 0.4) restricts displacement-guided matching to more reliable detections, yielding a higher IDF1 but lower Rcll; conversely, a lower θ_h (e.g., 0.3) admits more detections into Stage-I matching, which improves Rcll but at the cost of a lower IDF1 and more identity switches. Based on this analysis, we adopt $(\theta_h, \theta_l) = (0.35, 0.2)$ as our final configuration, as it achieves a favorable balance between strong identity consistency (74.7% IDF1) and high trajectory coverage (75.1% Rcll), leading to a competitive overall MOTA. This choice aligns with our goal of maintaining both tracking continuity and identity preservation under reliability-variant observations in satellite videos.

E. Discussion

While RAST shows strong robustness on satellite video MOT benchmarks, several challenging boundary cases remain.

First, prolonged full occlusion is still difficult for the current online framework. Although HDCAS improves fault tolerance through displacement-guided association and Kalman-filter-based inertial matching, reliable observations may be absent for multiple consecutive frames under long-duration occlusion, causing prediction uncertainty to accumulate. As a result, the corresponding track may be terminated by the online lifecycle management strategy, and the target may be reinitialized as a new trajectory after reappearance, leading to trajectory fragmentation or identity switches.

Second, the proposed framework may become less reliable under severe platform jitter or abrupt global motion disturbance. Since RAST relies on temporal feature differencing and learned displacement cues, strong background-induced pseudomotion may contaminate the motion prior in MGDFAM and reduce the reliability of displacement-guided association in HDCAS, especially for tiny targets with weak motion signatures.

These observations indicate that the current RAST framework is still limited in handling extreme long-term occlusion and severe global-motion interference. Future work will focus on incorporating longer-term identity recovery mechanisms and more robust global motion modeling to further improve tracking reliability in challenging satellite scenes.

V. CONCLUSION

This article presented RAST, an online and causal framework for MOT in satellite remote sensing videos from a reliability-aware spatiotemporal modeling perspective. By jointly addressing temporal representation drift, task interference between detection and displacement estimation, and reliability-variant association, RAST achieves robust tracking through a unified causal pipeline. Extensive experiments on SatVideoDT and SatMTB-MOT demonstrate that RAST consistently outperforms existing SOTA methods in tracking accuracy, trajectory continuity, and cross-category generalization. Future work will focus on improving long-term identity recovery under prolonged occlusion, developing more robust global motion modeling, and enhancing runtime efficiency for practical satellite video tracking.

REFERENCES

[1] J. Yang et al., "EHTracker: Toward fine-grained localization for satellite video target tracking," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5582–5599, 2025.

[2] F. Liu et al., "Remote sensing video tracking: Current status, challenges and future," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 14338–14367, 2025.

[3] L. Kong, Z. Yan, H. Shi, T. Zhang, and L. Wang, "LocaLock: Enhancing multi-object tracking in satellite videos via local feature matching," *Remote Sens.*, vol. 17, no. 3, p. 371, Jan. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.

[4] Z. Feng, X. Zhang, B. Zhou, M. Ren, and X. Zhi, "Moving vehicles tracking from satellite video data based on spatiotemporal high-order relation learning and reasoning," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 146, Feb. 2026, Art. no. 105015.

[5] X. Ying et al., "Infrared small target detection in satellite videos: A new dataset and a novel recurrent feature refinement framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5002818.

[6] R. Fan, J. Ye, H. Chen, Z. Huang, X. Wang, and W. Li, "SatSAM2: Motion-constrained video object tracking in satellite imagery using promptable SAM2 and Kalman priors," 2025, *arXiv:2511.18264*.

[7] R. Gao, J. Qi, and L. Wang, "Multiple object tracking as ID prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 27883–27893.

[8] Z. Pi et al., "Very low-resolution moving vehicle detection in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624517.

[9] F. Hu, P. Yang, J. Dou, and L. Dou, "Multiscale collaborative attention network for robust multiobject tracking in satellite video," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 28635–28647, 2025.

[10] J. Yu et al., "On-satellite implementation of real-time multi-object moving vehicle tracking with complex moving backgrounds," *Remote Sens.*, vol. 17, no. 3, p. 418, Jan. 2025.

[11] M. Zhao, S. Li, H. Wang, J. Yang, Y. Sun, and Y. Gu, "MP2Net: Mask propagation and motion prediction network for multiobject tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5617515.

[12] B. Lin, J. Zheng, C. Xue, L. Fu, Y. Li, and Q. Shen, "Motion-aware correlation filter-based object tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5604313.

[13] J. Hong, T. Wang, Y. Han, and T. Wei, "Multi-target tracking for satellite videos guided by spatial–temporal proximity and topological relationships," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614020.

[14] Z. Xiao, F. Xu, C. Liu, W. Yang, and G.-S. Xia, "SPAX: Fully sparse framework with hierarchical spatio-temporal fusion for moving object tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5600215.

[15] Y. Chen, Q. Yuan, Y. Xiao, Y. Tang, J. He, and T. Han, "STAR: A unified spatiotemporal fusion framework for satellite video object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631122.

[16] X. Zhou, V. Koltun, and P. Krähenbühl, "Tracking objects as points," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 474–490.

[17] T. Meinhardt, A. Kirillov, L. Leal-Taixe, and C. Feichtenhofer, "TrackFormer: Multi-object tracking with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8844–8854.

[18] W. Zhang, W. Deng, Z. Cui, J. Liu, and L. Jiao, "Object knowledge distillation for joint detection and tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701213.

[19] S.-K. Weng, C.-M. Kuo, and S.-K. Tu, "Video object tracking using adaptive Kalman filter," *J. Vis. Commun. Image Represent.*, vol. 17, no. 6, pp. 1190–1208, Dec. 2006.

[20] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3645–3649.

[21] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 3464–3468.

[22] Y. Zhang et al., "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 1–21.

[23] Y. Guo et al., "The first challenge on moving object detection and tracking in satellite videos: Methods and results," in *Proc. 26th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2022, pp. 4981–4988.

[24] S. Li et al., "A multitask benchmark dataset for satellite video: Object detection, tracking, and segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611021.

[25] L. Xu and Y. Huang, "Rethinking joint detection and embedding for multiobject tracking in multiscenario," *IEEE Trans. Ind. Informat.*, vol. 20, no. 6, pp. 8079–8088, Jun. 2024.

[26] Z. Liu, X. Wang, C. Wang, W. Liu, and X. Bai, "SparseTrack: Multi-object tracking by performing scene decomposition based on pseudo-depth," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4870–4882, May 2025.

[27] R. Luo, Z. Song, L. Ma, J. Wei, W. Yang, and M. Yang, "DiffusionTrack: Diffusion model for multi-object tracking," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 5, pp. 3991–3999.

[28] X. Yang et al., "Relation learning reasoning meets tiny object tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614516.

[29] W. Lu, B. Sun, S. Li, and X. Li, "FMA-Net: Flow-driven motion-aware network for multiobject tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5649515.

[30] L. Ren, W. Yin, W. Diao, K. Fu, and X. Sun, "Motion-guided multiobject tracking model for high-speed aerial objects in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5625815.

[31] Q. He, X. Sun, Z. Yan, B. Li, and K. Fu, "Multi-object tracking in satellite videos with graph-based multitask modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619513.

[32] H. Chen et al., "Multiple object tracking in satellite video with graph-based multiclue fusion tracker," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639914.

[33] L. Kong, Z. Yan, Y. Zhang, W. Diao, Z. Zhu, and L. Wang, "CFTracker: Multi-object tracking with cross-frame connections in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611214.

[34] J. Chen et al., "DeTracker: Motion-decoupled vehicle detection and tracking in unstabilized satellite videos," 2026, *arXiv:2601.09240*.

[35] H. Chen et al., "PIFTrack: Point-of-interest flows for multiobject tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5642314.

[36] Q. Wang et al., "TrackingMamba: Visual state space model for object tracking," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 16744–16754, 2024.

[37] J. Cao, J. Pang, X. Weng, R. Khirodkar, and K. Kitani, "Observation-centric SORT: Rethinking SORT for robust multi-object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9686–9696.

[38] Z. Li, D. Zhang, S. Wu, M. Song, and G. Chen, "Sampling-resilient multi-object tracking," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 4, pp. 3297–3305.

[39] N. Ballas, L. Yao, C. J. Pal, and A. Courville, "Delving deeper into convolutional networks for learning video representations," in *Proc. Int. Conf. Learn. Represent.*, 2016, pp. 1–10.

[40] Z. Xia, X. Pan, S. Song, L. E. Li, and G. Huang, "Vision transformer with deformable attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4794–4803.

[41] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[42] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[43] M. Zhao, S. Li, S. Xuan, L. Kou, S. Gong, and Z. Zhou, "SatSOT: A benchmark dataset for satellite video single object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617611.

[44] P. Bergmann, T. Meinhardt, and L. Leal-Taixe, "Tracking without bells and whistles," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 941–951.

[45] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the fairness of detection and re-identification in multiple object tracking," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3069–3087, Nov. 2021.

[46] J. Feng et al., "Cross-frame keypoint-based and spatial motion information-guided networks for moving vehicle detection and tracking in satellite videos," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 116–130, Jul. 2021.

[47] C. Xiao et al., "DSFNet: Dynamic and static fusion network for moving object detection in satellite videos," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[48] C. Xiao et al., "Highly efficient and unsupervised framework for moving object detection in satellite videos," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 11532–11539, Dec. 2024.

[49] A. Milan, L. Leal-Taixe, I. Reid, S. Roth, and K. Schindler, "MOT16: A benchmark for multi-object tracking," 2016, *arXiv:1603.00831*.

[50] F. Yu, D. Wang, E. Shelhamer, and T. Darrell, "Deep layer aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2403–2412.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:08:24 UTC from IEEE Xplore. Restrictions apply.